

Foreign Direct Investment and Gender Education Gap*

Rufei Guo Haiji Lin Xin Wang Chaoqun Zhan[†]

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Abstract

The gender education gap has been narrowing worldwide in recent decades. In this study, we present the first causal evidence of how Foreign Direct Investment (FDI) affects the gender education gap. By exploiting the quasi-exogenous deregulation of FDI entry in China upon its WTO accession, we find that prefectures that experienced greater FDI deregulation tend to have a lower gender education gap for the exposed cohorts. This finding is robust to various identification challenges. We identify two key mechanisms: a job-opportunity channel, where female-skill-intensive FDI shifts labor demand towards skilled females, incentivizing female educational investment; and a culture-spillover channel, whereby FDI diffuses gender-equal social norms, reducing son preference and promoting gender equality in education. In addition, FDI deregulation also has a persistent effect on gender equality by empowering women’s careers.

JEL Classification: F16, F66, I24, J16, J24

Key Words: Foreign Direct Investment; Gender Gap; Human Capital; Culture

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[†]Rufei Guo, Center for Economic Development Research, Center for Health Economics and Management, Wuhan University. Haiji Lin, Center for Health Economics and Management, Economics and Management School, Wuhan University. Xin Wang, Institute of New Structural Economics, Peking University. Chaoqun Zhan (Corresponding author), Department of Accountancy, Economics and Finance, Hong Kong Baptist University, email: cqzhan@hkbu.edu.hk.

1 Introduction

The narrowing gender gap in educational attainment represents a salient trend in human capital development over recent decades. Yet, the underlying forces behind this global convergence remain insufficiently understood. During the same period, the world has also experienced a rapid acceleration of economic globalization, with Foreign Direct Investment (FDI) playing a pivotal role ([Antràs and Yeaple, 2014](#)). This parallel development raises a natural question: Does the rapidly growing FDI contribute to closing the gender education gap? If so, the global diffusion of multinational firms and the widespread policy efforts by governments to attract FDI suggest that FDI may play a meaningful role in shaping global gender dynamics.¹

Although international trade has been extensively studied for its impact on human capital accumulation and gender education gap, the role of FDI remains comparatively underexplored. Unlike trade, FDI involves the direct operation of production facilities in host countries, bringing equipment, personnel, and management practices from abroad and engaging in face-to-face interactions with local individuals and firms. These differences suggest alternative channels through which FDI may affect educational outcomes by gender. For example, foreign firms are more likely to base their hiring decisions on industry-specific skill needs rather than prevailing social expectations in host countries, potentially opening up employment opportunities for women in sectors where domestic firms have historically overlooked female talent. In addition, foreign firms may transmit organizational norms shaped

¹Multinational Corporations (MNCs) account for about 80% of world trade and 25% of world GDP in 2010; and 6092 International Investment Agreements have been signed around the world to promote FDI ([UNCTAD, 2011, 2013](#)).

by their home-country institutions, which can spill over into workplace culture in the host country and gradually influence norms around women’s participation in the workforce. Recent studies document that such cultural transmission can support more inclusive practices and raise female employment in domestic firms ([Kodama, Javorcik, and Abe, 2018](#); [Tang and Zhang, 2021](#)).

We examine the causal relationship between FDI and gender education gap, exploring quasi-experimental variations of FDI deregulation in China. China provides an ideal setting to investigate this question for three reasons. First, China has experienced a rapid narrowing of the gender education gap in recent decades ([Rosenzweig and Zhang, 2013](#)). In 1990, 22-year-old men had, on average, 0.69 more years of schooling than 22-year-old women; by 2010, the gender gap had narrowed by 74 percent to 0.18 years.² Second, China greatly relaxed FDI entry regulations after its accession to the World Trade Organization (WTO), following commitments made in rounds of negotiations before the WTO accession. Around one-fourth of manufacturing industries experienced a relaxation of FDI entry regulation. To construct the regional exposure to FDI deregulation, we exploit the fact that multinational firms are more likely to set up affiliates in areas with industry clusters. Prefectures with a larger share of the national total in a deregulated industry are therefore more likely to attract FDI inflows in that sector following the regulatory changes. Consequently, differences in *ex ante* industrial structures across prefectures translate into varying degrees of exposure to FDI deregulation. Our methodology is analogous to the supply-push instrument in the literature on foreign aid ([Temple and Van de Sijpe, 2017](#)), which allocates a donor’s total aid to each

²Using the 2010 China Population Census, we calculate mean years of schooling by gender and birth year. Because individuals who have reached age 22 should have completed their college education, comparing years of schooling at age 22 captures schooling trends for the young generation. We assign each birth cohort to the calendar year in which it reaches age 22 by adding 22 to the birth year. Online Appendix Figure B1 presents the trends in educational attainment by gender.

recipient country based on their propensity to receive the aid flow. This allows us to exploit plausibly exogenous variations in the relaxation of FDI regulations across regions (Lu, Tao, and Zhu, 2017; Eppinger and Ma, 2024; Liu, Qiu, and Zhan, 2024).³ Third, the detailed individual, household, firm, industry, and region-level data in China allow us to explore the mechanisms through which FDI deregulation affects the gender gap in human capital in depth.

Specifically, we exploit individuals’ different exposure to the relaxation of FDI regulations at age 16, the “key exposure age” when individuals decide whether to pursue high school education or enter the workforce (Atkin, 2016; Li et al., 2019; Zhang and Zhou, 2023). Age 16 marks the completion of compulsory education in China, as the *Compulsory Education Law* mandates 9 years of education, and children typically start school at age 6 or 7. Additionally, according to China’s Labor Law, age 16 is the minimum legal working age. We focus on cohorts born in 1983–1988, who were aged 14–19 when the FDI regulations were relaxed in 2002. Individuals aged 14–16 in 2002 were on the edge of entering high school and are considered the “treated” cohorts, while those aged 17–19 had already made decisions regarding high school entrance and thus serve as the “untreated” cohorts. We utilize the variations in FDI deregulation across cohorts and regions to identify the causal effect of FDI on the gender gap in human capital.

We find that FDI deregulation significantly narrows the gender gap in high school attendance. Specifically, prefectures that experienced greater FDI deregulation tend to exhibit a lower gender education gap for the treated cohorts. This effect is economically sizable. According to our preferred estimate, a median increase in FDI deregulation raises the like-

³Each region is a prefecture. We have data on 328 prefectures. We interchangeably use “region” and “prefecture” to refer to the same geographical unit.

likelihood of high school attendance for females relative to that of males by 0.78 percentage points, representing 23.9% of the gap of the untreated cohorts. While the reduction in the gender education gap is similar in magnitude for both rural and urban areas, the narrowing is more clearly attributable to increases in female educational attainment in rural regions, which are generally associated with stronger gender bias. Consistent with this pattern, we also find that the estimated impact is larger and more statistically significant in areas with higher genealogy density, which serves as a proxy for stronger pre-existing gender bias.

To address concerns about endogeneity, our baseline regressions incorporate a rich set of controls. First, the main results are obtained after controlling for interactions between cohort dummies and various initial prefecture characteristics, such as the employment share of agriculture and manufacturing, to account for the potential effect stemming from the dynamics of manufacturing and service sectors. Second, we control for contemporaneous trade reforms, including the import tariff cuts in China and the reductions in US trade policy uncertainty toward China, to isolate the effects of FDI deregulation from broader trade liberalization. We further assess the credibility of the identification strategy through a series of robustness checks. We first verify the validity of the parallel trends assumption by showing that regions with different levels of exposure to FDI deregulation exhibit similar pre-trends in educational outcomes prior to the policy change. In addition, we consider other factors that may affect the estimated relationship, including vertical FDI linkages, spatial spillovers, regional development levels, and selective migration. Finally, we address key identification concerns specific to shift-share designs. Overall, these exercises reinforce the credibility of our identification strategy and the robustness of the estimated effects.

Our mechanism tests reveal two channels through which FDI shapes the gender edu-

cation gap: a job-opportunity effect and a culture-spillover effect. The literature has well documented that globalization creates job opportunities for developing countries ([Atkin, 2016](#); [Li et al., 2019](#)). To capture gender-specific variation in skill demand, we follow [Li \(2021\)](#) and construct a female-skill-intensive FDI shock by interacting industry-level FDI deregulation with the relative importance of female-oriented skills across industries. We also include a reverse measure in the regression to capture male-skill-intensive FDI shocks. Our results show that in regions with greater exposure to female-skill-intensive FDI, the female-to-male employment ratio increases significantly among high-skilled workers, particularly those aged 1630, while no comparable effects are observed for low-skilled workers. In comparison, male-skill-intensive FDI shocks reduce the female-to-male employment ratio among high-skilled workers.

In addition to the job-opportunity channel, we find evidence supporting FDI’s culture-spillover effect. As noted earlier, the impact of FDI on narrowing the gender education gap is more pronounced in regions where gender bias was initially higher. This pattern suggests that increased job opportunities alone do not fully explain FDI’s influence on the gender education gap. Instead, the evidence points to a cultural mechanism, whereby FDI diffuses gender-equal culture to suppress pre-existing gender bias against females ([Kodama, Javorcik, and Abe, 2018](#); [Tang and Zhang, 2021](#)). To provide more direct evidence for this cultural channel, we examine FDI’s impact on the sex ratio at birth, a widely used measure of son preference ([Jayachandran, 2015](#)). Our analysis reveals that FDI deregulation reduces the male-biased sex ratio. Consistent with our findings on the gender education gap, we also find that this effect is significant only in regions where cultural practices were gender unequal prior to deregulation.

We further find that FDI has a persistent effect on gender inequality by empowering women’s careers in the longer run. We examine the labor market performances of our sample cohorts eight years after the shock, by which time most individuals have completed their college education. While females and males are equally capable of finding jobs eight years after the shock, FDI deregulation increases the likelihood of females securing white-collar jobs relative to males. Complementing [Tang and Zhang \(2021\)](#)’s finding that MNEs tend to hire more female managers, our results indicate a longer-run empowerment effect of FDI on women’s careers through human capital accumulation.

This study contributes centrally to the extensive debate concerning the effects of globalization on human capital accumulation. While much of this literature has focused on the influence of trade, particularly via channels including labor market dynamics (see [Atkin, 2016](#); [Greenland and Lopresti, 2016](#); [Blanchard and Olney, 2017](#); [Li, 2018](#); [Li et al., 2019](#)) and household income and intra-household resource allocation (see [Edmonds, Pavcnik, and Topalova, 2010](#); [Dai, Huang, and Zhang, 2021](#)), we extend this discussion by investigating impacts of FDI on human capital accumulation. While prior research has primarily emphasized employment-based channels linking globalization to gender outcomes, our study highlights that these mechanisms alone may be insufficient. We offer the first causal evidence that FDI plays a role in narrowing the gender education gap, operating through both labor market opportunities and a complementary cultural channel that influences gender norms. This expands our understanding of the mechanisms through which globalization influences human capital formation, shedding light on previously overlooked channels.

Our paper is also closely related to studies on the impacts of globalization on gender inequality. The literature has extensively examined the effects of trade on gender inequality in

the labor market ([Sauré and Zoabi, 2014](#); [Juhn, Ujhelyi, and Villegas-Sanchez, 2014](#); [Gaddis and Pieters, 2017](#); [Kis-Katos, Pieters, and Sparrow, 2018](#); [Bøler, Javorcik, and Ulltveit-Moe, 2018](#); [Yu et al., 2021](#); [Wang, Kis-Katos, and Zhou, 2024](#); [Keller and Utar, 2022](#); [He et al., 2025](#); [Luo, Zeng, and Zhao, 2025](#)). For example, [Keller and Utar \(2022\)](#) shows that even a gender-neutral adverse trade shock results in significant gender gaps in the labor market in the long run, driven by the female biological clock. A growing but limited body of literature has examined the impacts of FDI on gender inequality in the labor market. [Kodama, Javorcik, and Abe \(2018\)](#), for example, find that MNEs tend to employ proportionately more women and more female managers than domestic firms in Japan. Similar results are obtained by [Choi and Greaney \(2022\)](#) for Korea. Our paper is closely related to [Tang and Zhang \(2021\)](#), who conclude that MNEs transfer gender-equal employment practices and gender-equal culture across borders in China. Differing from their focus on the labor market, we examine the impacts of FDI on gender gap in human capital investment. Our findings imply that the global diffusion of gender-equal culture, possibly facilitated by the deregulation of transnational flows of capital, acts as an important determinant of worldwide changes in the gender education gap.

This paper is organized as follows. Section [2](#) describes the data we use and introduces our empirical strategy. Section [3](#) reports the baseline regression results and checks their validity and robustness. We explore the mechanisms in Section [4](#) and discuss the longer-term effect in Section [5](#). Section [6](#) concludes the paper.

2 Data and Empirical Strategy

2.1 Data

Industry-level FDI Regulation Data

During our sample period, China imposed strict regulations on FDI inflows, with the most notable being the restriction on FDI entries. Specifically, in June 1995, the Chinese government promulgated the Catalogue for the Guidance of Foreign Investment Industries (henceforth, the Catalogue), which classifies all industries/products into four categories, depending on whether FDI is (i) supported, (ii) permitted, (iii) restricted, or (iv) prohibited. Since the first promulgation, the Catalogue has undergone multiple revisions. The first revision occurred in 1997, which was minor. A subsequent and major revision was announced in March 2002 soon after China’s accession to the WTO in December 2001. Minor revisions were also implemented in 2004, 2007, and 2011. We exploit the revision of the Catalogue in 2002.⁴ This revision is arguably exogenous because it follows the commitments of the “Agreement on Trade-Related Investment Measures” for China’s WTO accession, which was formed during rounds of multilateral negotiations that started long before 2002 (Lu, Tao, and Zhu, 2017).

We compare the 1997 and 2002 versions of the Catalogue to construct an industry-level indicator of FDI deregulation. Following Lu, Tao, and Zhu (2017), we first examine product-level changes in FDI entry regulation in the 2002 Catalogue compared with the 1997 Catalogue. We classify all products into three groups according to the changes: “more

⁴We do not use the revisions of 1997, 2004 and 2007 because they involve only a few modifications. For example, relative to the 1995 version, the 1997 revision relaxed restrictions in only four industries, shifting their classification from “permitted” to “encouraged”. Excluding these industries from our analysis does not affect our results.

welcome”, “more restrictive”, and “no change”. We then define FDI policy change for each 4-digit Chinese Industry Classification (CIC) industry, which belongs to one of the following four categories: (i) FDI relaxed industry, in which products are either “more welcome” or “no change” and at least one product is “more welcome”; (ii) no-change industry, in which all products experience no change in FDI regulation in 2002; (iii) FDI restricted industry, in which products became either “more restrictive” or “no change”, and at least one product is “more restrictive”; and (iv) mixed industry, in which some products are “more welcome” but some are “more restrictive”. Based on the above classification, we find that the last two groups together contain only 15 industries. We exclude the last two groups from our data set. As a result, we obtain 119 FDI relaxed industries and 302 no-change industries. This suggests that the revision in 2002 provides substantial variation for the empirical analysis, as more than 25% of the manufacturing industries experienced relaxation in FDI entry in this reform. We combine industry-level FDI deregulation with the region’s initial labor share in that industry to construct regional FDI shocks, which are discussed in detail in Section 2.2.

Individual-level Population Census Data

Our primary data source of outcome variables is the 2005 and 2010 waves of the China population (mini-)census, which is carried out by China’s National Bureau of Statistics (NBS). We have access to a 20% sample of the 2005 mini-census that surveyed approximately 1% of the Chinese population and a 0.36% sample of the 2010 census that covers the whole population. The 2005 census data contain 2,585,481 observations, and the 2010 census data contain 4,803,589 observations. The census data contains detailed individual information on age, ethnicity, education, employment, and marriage. We derive individuals’ demographic

characteristics, education attainments, and employment information from the census data.

Firm-level Balance Sheet Data

We construct the employment weight in our main measure of prefecture-level FDI shock based on the Above-Scale Industrial Firms Panel (ASIFP), which is available for 1998–2007. The ASIFP is also provided by NBS. It covers all state-owned enterprises (SOEs) and non-SOEs with annual sales of at least 5 million RMB from all 31 provinces and directly administered municipalities in China. We are confined to firms from manufacturing industries. The ASIFP contains detailed information on each firm, including the firm’s official name, industry, location, ownership, and employment. We clean the ASIFP data following [Brandt, Van Biesebroeck, and Zhang \(2012\)](#).

Supplementary Data

To control for other contemporaneous reforms, we also include sudden and drastic import tariff reductions and trade policy uncertainty reductions during China’s accession to the WTO. We obtain tariff data from the World Integrated Trade Solutions (WITS) and trade policy uncertainty data from [Pierce and Schott \(2016\)](#). We also use the 1990 China population census data to construct pre-determined regional characteristics and the 2000 China population census in robustness tests. The information on firms’ employment across gender and education level in the 2004 Economic Census by NBS is employed to calculate gender-related variables between FDI and domestic firms. Furthermore, we utilize the Occupational Information Network (O*NET) database and employment data from U.S. Bureau of Labor Statistics to quantify the intensity of gender-oriented skills across industries.

2.2 Empirical Strategy

Regional FDI Shocks

We utilize the regional variation in FDI shocks to identify its effect on the gender education gap. As documented in [Lu, Tao, and Zhu \(2017\)](#), industries that experienced FDI deregulation received more FDI inflows after the reform. We measure the regional exposure to FDI deregulation as:

$$FDI_r = \sum_k \frac{Emp_{kr0}}{Emp_{k0}} \cdot Treatment_k, \quad (1)$$

where Emp_{kr0} and Emp_{k0} are the total number of individuals employed in industry k of region r and of the whole economy in the base period 2000, respectively, based on data from ASIFP; $Treatment_k$ is a dummy indicating whether industry k experienced FDI deregulation in 2002. We define the region at the prefecture level in China.

Our measure of regional exposure to FDI deregulation in Eq. (1) is the weighted sum of industry-specific FDI deregulation intensity, with region r 's importance as a producer in industry k serving as the weights. Multinational firms are inclined to establish affiliates in locations with industry clusters, which may possess natural or geographical comparative advantages, larger market potentials, or better access to suppliers for that particular industry's development. Our measurement exploits industry-specific changes in FDI regulations and variations in the propensity of each location to receive FDI inflows into China for each industry. This method is analogous to the supply-push instrument used in the literature on foreign aid ([Temple and Van de Sijpe, 2017](#); [Nunn and Qian, 2014](#); [Dreher et al., 2021](#)), which utilize the interaction of year-to-year changes in the supply of aid from each donor

country and the likelihood that each recipient country receives a smaller or larger share of these funds.

Figure B2 in the Online Appendix plots the spatial distribution of the regional FDI deregulation intensity. In general, we observe higher intensities with large variations across prefectures in eastern and central China and lower intensities with small variations in the western part. To validate that this measure captures the extent of FDI shock at the regional level, we examine whether FDI_r changes FDI activities at the regional level. We do find that regions with larger FDI_r receive more FDI activities after the reform (Online Appendix Table B1).

Schooling Decision at Age 16

Following [Atkin \(2016\)](#), [Li et al. \(2019\)](#) and [Zhang and Zhou \(2023\)](#), our identification strategy exploits differences in the exposure to FDI deregulation across cohorts around age 16 within the same prefecture. We focus on age 16 as the “key exposure age” for several reasons. First, the *Compulsory Education Law* enacted in 1986 mandates the completion of middle school (ninth grade) for all individuals. Upon finishing middle school, most people are 16 years old. Second, age 16 is the minimum legal working age. Therefore, age cohort 16 is the pivotal cohort when individuals must decide between pursuing further education or entering the workforce. At this juncture, students and parents trade off the benefits of higher future earnings by attending high school with the current wage premium of working. Therefore, attendance of high school at age 16 is a sensitive margin of education that could potentially be affected by FDI deregulation.

Specifically, we include people born in 1983–1988 who were aged 14–19 at the relaxation

of FDI regulations in 2002 in both the 2005 mini-census and 2010 census. People aged 14–16 in 2002 were about to enter high school, while people aged 17–19 had already made decisions on high school entrance. Therefore, people aged 14–16 in 2002 are the treated cohorts, and people aged 17–19 are the untreated cohorts. For these 1983–1988 cohorts, we can observe their decisions on high school education in both the 2005 and 2010 censuses.⁵ To increase statistical power, we use a pooled sample of the 1983–1988 cohorts from both the 2005 and 2010 censuses. Results are similar if we use 2005 or 2010 census alone, albeit with lower statistical power.

We obtain 343,087 males and 355,320 females in our sample (Table 1). Approximately 90% of the sample is of Han ethnicity, and 22% is from urban areas. Age, ethnicity, and urban status differ little by gender. Most of our sample people attended middle school: The rate of middle-school attendance was 91%. High school attendance is much less common. Approximately 41% of males and 37% of females had ever attended high school. Most people have not attended college.

Regression Specification

We utilize the plausibly exogenous relaxation of FDI regulation following China’s WTO accession to investigate the causal effect of FDI deregulation on the gender education gap. Specifically, we estimate the gender-interactive model using the following specification:

$$Y_{irc} = \beta_1 FDI_r \cdot Age16_c + \beta_2 FDI_r \cdot Age16_c \cdot Female_i + \mathbf{X}'_{irc} \mathbf{\Gamma} + \lambda_{gr} + \lambda_{gpc} + \varepsilon_i, \quad (2)$$

⁵We can observe the decisions of high-school entrance of the 1983–1988 cohorts in the 2010 census as they age at 22–27 in 2010. We also observe the decisions of these cohorts in the 2005 mini-census. When people born in 1988 (the youngest cohort) were observed in the 2005 mini-census, they aged 17 years old, which is above the usual age of 16 for high school entrance.

where i indexes for individual, r for region, c for cohort (birth year), g for gender, and p for province. Y_{irc} is an indicator that is equal to one if individual i of cohort c in region r has ever attended high school, and zero otherwise. FDI_r denotes the intensity of FDI deregulation in region r , as defined in the previous section;⁶ The variable $Age16_c$ is an indicator that is equal to one if individual i of cohort c is no more than 16 years old in 2002 and otherwise zero. $Female_i$ is an indicator of being female. We also allow other control variables in $\mathbf{X}_{irc}$ to have separate coefficients for each gender. The vector $\mathbf{X}_{irc}$ includes two sets of covariates. The first is a set of individual controls, including whether the individual is of Han ethnicity and whether he or she holds an urban hukou. The other is a set of interactions between people's cohort indicators and pre-determined regional characteristics to allow education attainments to vary systematically by these characteristics. These pre-determined regional characteristics include the average years of schooling, the share of agricultural employment, and the share of manufacturing employment for people aged 16–60 in the 1990 China population census. The inclusion of the last two shares also accounts for the possible effects of the variation in the service sector in each region for each cohort (Ngai and Petrongolo, 2017; Petrongolo and Ronchi, 2020). We further include the gender-specific prefecture fixed effects λ_{gr} , and λ_{gpc} gender-specific province-by-cohort fixed effects. All other two-term interactions have been included in these fixed effects. We cluster the standard error ε_i at the prefecture level to account for serial correlation problems following Bertrand, Duflo, and Mullainathan (2004).

The coefficient β_2 is a triple-difference estimator that captures the effect of FDI deregulation on female high school attendance relative to male high school attendance. The first difference compares the high-school attendance rate of the younger (treated) cohorts versus

⁶We use individuals' hukou registration information to determine which prefecture they belong to, because most people enroll in high schools in the prefecture of hukou registration.

that of the older (untreated) cohorts. The younger cohorts, who were 14–16 years old at the relaxation of the FDI regulations in 2002, were yet to make their decision of high-school attendance; while the older cohorts, who were 17–19 years old in 2002, had passed the usual age of high-school entrance. The second difference compares the high-school attendance rate between regions with high intensity of FDI deregulation and regions with low deregulation intensity. The third difference compares between females and males.⁷

Contemporaneous Reforms

Contemporaneous reforms might contaminate the validity of our DID estimation. If other reforms introduced around the time of China’s WTO accession systematically correlate with FDI deregulation across regions, then the effects of those reforms would also be picked up in our DID estimates. Several reforms took place in the early 2000s. We discuss how to control for their possible confounding effects.

One of the most important contemporaneous reforms is trade liberalization in China. After its accession into the WTO, China experienced a drastic and sudden drop in import tariffs across industries. Regions with different industrial structures are therefore affected differently. The literature has extensively investigated the effects of trade liberalization on human capital and gender inequality. [Fieler and Harrison \(2023\)](#), for example, show that tariff cuts are associated with employment switches to skill-intensive sectors in China, based on which we might expect further impacts on human capital. In addition, since the tariff and FDI reforms are both commitments of China on the entry of the WTO, they might be correlated. To control for its possible confounding effect, we include $Tariff_r \cdot Age16_c$ in our

⁷All other single terms and two-term interactions are included but not reported.

regression, where $Tariff_r$ is constructed with industry-level tariffs in 2001, $T01_k$, according to a formula similar to Eq. (1): $Tariff_r = \sum_k \frac{Emp_{kr0}}{Emp_{k0}} \cdot T01_k$.⁸

The second is trade policy uncertainty reduction in the US. After China's accession to the WTO, the US granted China permanent normal trade relations (PNTR) which resolved tariff uncertainty. [Pierce and Schott \(2016\)](#) show that industries with larger reductions in uncertainty export more to the US after the reform. If the uncertainty reduction across regions is systematically correlated with the effect of FDI reform, our estimate might pick up the effect from trade policy uncertainty reduction. To isolate the effect of FDI reform from that of uncertainty reduction, we further control for $PNTR_r \cdot Age16_c$ in our baseline regression, where $PNTR_r$ is constructed with industry-level column 2 tariff, $col2_k$, according to $PNTR_r = \sum_k \frac{Emp_{kr0}}{Emp_{k0}} \cdot col2_k$.⁹

The third is China's higher education expansion. In 1999, China expanded college admissions by 40%, and the high expansion rate lasted until college enrollment quadrupled by 2005 ([Che and Zhang, 2018](#)). An expansion of college education would increase the demand for high school education, which is necessary for college entrance. College expansion is mainly through enlarged admissions by existing colleges instead of establishing new ones. Colleges allocate admission quotas to each province with more quotas for local students, so a large proportion of quotas is allocated to the province where the college is located. High school students take the college entrance exam to compete for scarce admission quotas within each province. Therefore, provinces with more colleges are expected to experience a larger

⁸To avoid endogeneity in the process of tariff reduction, we follow [Liu, Qiu, and Zhan \(2019\)](#) and use the initial tariff level in 2001 instead of the actual tariff reduction. The initial tariff level is highly positively correlated with actual tariff reductions in China, and therefore can represent the treatment intensities.

⁹Before granting PNTR to China, the U.S. congress annually reviews and decides whether to approve normal trade relations (NTR) with China. If the NTR with China was not approved, tariffs would be set back to very high tariff rates, called column 2 tariffs, which were determined decades ago. Therefore, the uncertainty reduction due to PNTR status is higher for industries with higher column 2 tariffs

increase in quotas after the reform, which may influence people’s decisions to attend high school. Nevertheless, this possible confounding effect has been teased out by the province-by-cohort fixed effect λ_{pc} in our regression, which absorbs all time-varying province-level shocks.

3 Primary Findings

3.1 Baseline and Heterogeneity Analysis

Baseline Estimates

Column (1) of Table 2 reports the baseline estimates derived from Eq. (2). Our specification employs a comprehensive set of controls to address potential identification challenges. First, we account for contemporaneous policy reforms coinciding with China’s WTO accession, including trade liberalization stemming from import tariff reductions and the elimination of trade policy uncertainty with the US, thus mitigating concerns that our estimates might capture these confounding effects.

Second, we incorporate two categories of covariates. At the individual level, we control for hukou status and ethnicity as key determinants of educational attainment in China. At the regional level, we include predetermined characteristics such as average years of schooling and the share of agricultural and manufacturing employment among working-age individuals. We interact these regional factors with cohort dummies to allow educational outcomes to vary systematically across cohorts based on these economic characteristics. Third, we include prefecture fixed effects to capture time-invariant prefecture characteristics, and province-by-cohort fixed effects to absorb province-specific trends in educational attainment.

Notably, our baseline results indicate that the gender-differential effect of FDI relaxation on high school attendance is positive and statistically significant at the 1% level, suggesting that FDI deregulation substantially contributes to narrowing the gender education gap in China. The estimated effect is economically sizable. The median value of the FDI deregulation variable is 0.140. According to the estimate in column (1), a median increase in FDI deregulation raises the likelihood of high school attendance of females relative to that of males by $0.056 \times 0.140 \times 100 = 0.78$ percentage points. For cohorts born between 1983 and 1985, who were above 16 years old in 2002, the gender gap in the likelihood of high school attendance is 3.26 percentage points. Our estimate suggests that FDI deregulation narrowed the gap by $\frac{0.78}{3.26} = 23.9\%$. This narrowing is driven by a relative decrease in high school attendance among males in the exposed cohorts.

Heterogeneity Analysis: Hukou Status

We first divide the full sample into rural and urban subsamples. China’s hukou system constitutes a core institutional framework that governs internal population registration of Chinese citizens ([Chan and Zhang, 1999](#); [Wing Chan and Buckingham, 2008](#)). Children inherit their hukou classification—either urban or rural—from their parents, and this in turn, critically determines their eligibility to access public education resources, which provides the majority of secondary education in China ([Chan, 2009](#)).

Columns (2) and (3) in Table 2 show the estimates of Eq. (2) using subsamples of people with rural hukou and urban hukou, respectively. The deregulation of FDI reduces the gender education gap in both rural and urban subsamples by similar magnitudes, indicating that its effects are fairly consistent across socioeconomic strata. In particular, within the rural

hukou subsample, which constitutes approximately 78 percent of our baseline estimation sample, the convergence in gender educational attainment following FDI deregulation is predominantly attributable to a substantial rise in female high school attendance.

Heterogeneity Analysis: Gender-unequal Culture

We next investigate heterogeneous effects through the lens of gender-unequal culture. As the predominant culture lineage in the Chinese society, Confucian culture embodies gender-biased values and beliefs. Along the social hierarchy, Confucianism entrenches systematic gender bias against females ([Jayachandran, 2015](#); [Zhang, 2019](#)).¹⁰ Additionally, Confucian culture had been enforced by prescribed rituals and code of conduct ([Ebrey, 1991](#)). For example, in ancestor-worship ceremonies, the rituals ascribe male members much more superior status than female members. The Confucian code of conduct was recorded in genealogies, books that keep track of a lineage’s male members. The compilation of genealogies reflect the strength of Confucian culture. We use the density of genealogies to measure the strength of Confucian culture, following [Greif and Tabellini \(2017\)](#), [Zhang \(2019, 2020\)](#) and [Chen, Ma, and Sinclair \(2022\)](#).¹¹

As shown in column (4) and (5) of Table 2, we report the baseline analysis for prefectures with above- or below-median genealogy density. In regions with above-median genealogy density, FDI deregulation significantly closes the gender gap in high school attendance. In

¹⁰For example, Confucian classics had emphasized the importance of the “three obedience” for females: obedience to her father, to her husband, and even to her adult son. Using modern data, [Zhang \(2019\)](#) finds that a stronger Confucian culture predicts a more male-biased sex ratio at birth, a strong indicator of son preference ([Jayachandran, 2015](#)).

¹¹We obtained the genealogy density data from the Comprehensive Catalog on Chinese Genealogy (zhongguo jiapu zongmu) compiled by the [Shanghai Library \(2009\)](#), and followed [Chen, Ma, and Sinclair \(2022\)](#) to compute genealogy density. Specifically, we employ the number of genealogies compiled between 1980 and 2009 as a proxy for the density of Confucian clans in the modern era. We construct the density measure by dividing the genealogy count by the regional population size, adding one to account for zero values, and then taking the natural logarithm.

contrast, in regions with below-median genealogy density, where the culture norm tends to be gender-equal, FDI deregulation has no effect on the gender education gap. It appears that our baseline results are driven by regions where the pre-existing culture norm is gender-unequal.¹²

Event Study

The validity of our estimates relies on comparable trends across regions with different treatment intensities. To examine parallel pre-trends and dynamic effects across cohorts, we extend the gender-interactive Eq. (2) to an event-study specification:

$$Y_i = \sum_{a \neq 17} \beta_{1a} FDI_r \cdot \mathbf{I}[c = a] + \sum_{a \neq 17} \beta_{2a} FDI_r \cdot \mathbf{I}[c = a] \cdot Female_i + \mathbf{X}'_{irc} \mathbf{\Gamma} + \lambda_{gr} + \lambda_{gpc} + \varepsilon_i, \quad (3)$$

where $\mathbf{I}[c = a]$ is an indicator variable that is equal to one if individual i is a year old in the year of FDI deregulation.¹³ We estimate the effects for ages 14–19, excluding age 17 as the reference group. The coefficient β_{2a} captures the effect of FDI deregulation on female high school attendance relative to that of males for people of age a , with age 17 as the excluded reference group. Note that we allow all controls in $\mathbf{X}_{irc}$ to have gender-specific coefficients.

Figure 1 plots the estimates of β_{2a} in Eq. (3). FDI deregulation does not affect the gender difference in high school attendance for people who were 18 or 19 years old in the year of FDI deregulation; in contrast, FDI deregulation increases female high school attendance relative

¹²We note that subsample analysis of gender-unequal culture may capture potential confounders associated with pre-existing regional differences in gender-unequal norms. For instance, regions with a high intensity of gender-unequal culture might also exhibit a larger pre-existing gender education gap or differ in levels of economic development and private enterprise activity, as discussed in Zhang (2020). However, our robustness checks in Appendix A.2 demonstrate that the heterogeneous effects associated with pre-existing gender-unequal cultural norms remain consistent after accounting for these potential concerns.

¹³Cohort (birth year) has a one-to-one correspondence to age in 2002. The set of cohort dummies I_c is numerically equivalent to the set of age dummies $\mathbf{I}[c = a]$.

to males for people who were 16 years old or younger. The event-study estimates consistently show parallel pre-trends for older cohorts and further justify our choice of age 16 as the “key exposure age” for high school attendance.

3.2 Validity and Robustness

Vertical Linkages

We have shown that prefectures that are more directly exposed to FDI tend to have a narrowing gender education gap. However, FDI deregulation could also potentially affect the educational choices of residents in this region through vertical linkages. For example, if many FDIs enter the upstream of the car industry and reduce the input cost for the car industry, then regions specializing in the car industry will benefit more, even if these upstream FDIs are located in other regions. The boom of the car industry can in turn affect the educational choices of local people. We follow [Wang et al. \(2018\)](#) to construct exposures to upstream and downstream FDI reforms to assess whether our results are driven by these vertical linkages.

Specifically, we first calculate the upstream FDI shock for industry k as:

$$Treatment_{up_k} = \sum_{l \neq k} \frac{X_{lk}}{Y_k} \cdot Treatment_l,$$

where X_{lk} is the total value of industry l ’s product (as the upstream) used in industry k (as the downstream), and Y_k is the total output of industry k . Intuitively, the upstream FDI shock for industry k , $Treatment_{up_k}$, is larger when industry k ’s important upstream supplier industries experienced FDI relaxation. Then, we obtain the upstream FDI shock at

the regional level as:

$$FDIup_r = \sum_k \frac{Emp_{kr0}}{Emp_{k0}} \cdot Treatmentup_k,$$

Analogously, the downstream FDI shock at the regional level is:

$$FDIdown_r = \sum_k \frac{Emp_{kr0}}{Emp_{k0}} \cdot Treatmentdown_k,$$

where $Treatmentdown_k = \sum_{l \neq k} \frac{X_{kl}}{Y_k} \cdot Treatment_l$ measures the downstream FDI shock at the sector level. The results in column (1) of Table 3 show that our results are robust to FDI vertical linkages.

Spillover Effects from Other Regions

Our identification strategy relies on the assumption that there are no spillover effects of shocks from other regions on the gender education gap of the local region. For example, if a coastal region in southern China such as Shenzhen experienced a large shock in FDI entry, will that confound the difference in educational choices of males and females in other regions?

We adopt two approaches to investigate these non-local spillover effects. First, we construct a weighted average FDI shock of the neighboring regions according to $FDIneighbor_r = \sum_{j \in Neighbor_r} \frac{1}{N_r} FDI_j$, where $Neighbor_r$ denotes the set of prefectures sharing common borders with r , and N_r is the total number of neighboring prefectures of r . Alternatively, we also construct an average FDI shock from all other regions weighted by the inverse of distance: $FDIother_r = \sum_{j \neq r} \frac{1}{d_{rj}} FDI_j$, where d_{rj} is the distance between prefectures j and r . We incorporate the interaction of the constructed non-local shock with $Age16_c$ and $Female_i$ to

control for their potential effects on the gender education gap.

Columns (2) and (3) of Table 3 report the results incorporating the interactions of $FDI_{neighbor_r}$ and FDI_{other_r} , respectively. In both regressions, local FDI shock increases female education relative to male education, and the effects are statistically significant and quantitatively stable, indicating that our results are not driven by shocks from non-local regions. Additionally, we do not detect a significant effect of FDI shocks from other regions on local education.

Migration

If individuals are highly mobile across prefectures, FDI may affect cohorts' educational choices without its consequences being identifiable at the regional level. Despite the high migration cost (Tombe and Zhu, 2019), we observe large-scale migration flows in China following WTO accession (Facchini et al., 2019). Fortunately, this concern is not severe in our context since we are investigating the educational decisions of teenagers aged around 16, who mainly enroll in high schools in the prefecture of hukou registration. Nevertheless, some teenagers may migrate to seek job opportunities in other prefectures. We further assess whether our results are robust after taking care of migration.

First, we re-estimate Eq. (2) with a subsample excluding migrants. We define an individual as a migrant if he or she lived in a prefecture that is different from the prefecture of hukou registration.¹⁴ Our definition of migrants follows Facchini et al. (2019) and captures cross-prefecture migration. People who migrate within a prefecture, e.g., from rural counties to urban districts in the same prefecture, are not treated as migrants. It is also common

¹⁴Recall that we define an individual's region according to the hukou registration information.

for teenagers born in rural counties to attend high schools in urban districts of the same prefecture. Because the FDI shock is defined at the prefecture level, within-prefecture migration does not threaten our identification. Column (4) in Table 3 reports the results using the subsample of non-migrants. Our key estimate remains essentially unchanged, suggesting that our results are not driven by migrants.

In addition, we examine whether FDI deregulation directly affects people’s migration decisions. Specifically, we check whether regions with larger FDI shocks tend to have more in-migration or out-migration. We define the in-migration rate of prefecture r as the share of immigrants in the residential population of the prefecture for people at ages 16–60 in prefecture r . Analogously, the out-migration rate of prefecture r is the share of people who migrated to other prefectures in the population who registered hukou in prefecture r for people at ages 16–60. We obtain the prefecture-level in-migration rate and out-migration rate for 2000, 2005, and 2010 using China population censuses, which record detailed information on migration history. We use the region-year specification to test whether the FDI shock affects migration:

$$M_{rt} = \gamma_1 FDI_r \cdot Post_t + \mathbf{Z}'_{rt} \mathbf{\Gamma} + \lambda_r + \lambda_{pt} + \varepsilon_{rt} \quad (4)$$

where M_{rt} is the in-migration rate or out-migration rate in prefecture r and year t ; $Post_t$ is the indicator for the reform and is equal to one from 2002 onward; and $\mathbf{Z}_{rt}$ includes the share of Han, the share of males, and interactions of year dummies with pre-determined regional characteristics as defined before.

Online Appendix Table B2 presents the results. Evidently, FDI relaxation does not have statistically significant effects on either the in-migration or out-migration rate for both

males and females. The estimates are quantitatively small as well. Local FDI shocks do not significantly induce migrations across regions. Overall, the results further confirm that migration is not a major concern in our study.

Differential Trends by Initial Development Level

Another concern is that higher-income regions are likely to attract more FDI entries following deregulation and tend to have a more rapid narrowing of the gender education gap compared with less developed regions. Globally, the gender education gap first closes in countries with higher incomes ([Bertocchi and Bozzano, 2020](#)). If income does correlate with the trend of the gender education gap, directly regressing educational choices on FDI deregulation might capture spurious causality. In our context, we have controlled for province-by-cohort fixed effects, which have controlled for the heterogeneity in economic developments across provinces between cohorts. In addition, we have also included the interactions between cohort dummies with the prefecture’s share of manufacturing employment for people aged 16–60 in 1990, which have partly reflected the economic conditions between different prefectures.

We further explicitly address the economic development concern by controlling for the GDP per capita for each prefecture. To alleviate the endogeneity concern that FDI deregulation might affect GDP per capita, we control for the interactions of cohort dummies with the prefecture’s GDP per capita in 2000. Column (5) of Table 3 show the results. The effect remains positively significant and quantitatively similar. In column (6), we further check the robustness by replacing GDP per capita with the average nightlight intensity of each prefecture in 2000.¹⁵ The results are also robust. In sum, our finding is not a spurious

¹⁵The average nightlight strength is measured as the simple average of all pixel values of satellite photos within the boundary of each prefecture during the night over one year and obtained from National Oceanic and Atmospheric Administration. For each pixel, a digital number ranging from 0 to 63 is assigned that represents the intensity of the

causality driven by the economic development level.

Other Education Outcomes

In our baseline specification, we focus on high school attendance, for which the “key exposure age” is 16. High school attendance is the sensitive margin of adjustment to trade liberalization (Atkin, 2016; Li et al., 2019). We further examine middle school and college attendance. For middle school attendance, we use the 1986–1991 cohorts and use 13 as the “key exposure age”. For college attendance, we use the 1980–1985 cohorts and use 19 as the “key exposure age”.¹⁶

Online Appendix Table B3 shows the effects of FDI deregulation on middle school and college attendance. The point estimates are small in magnitude and statistically insignificant across all specifications (columns (1)–(6)), suggesting that FDI deregulation does not meaningfully affect educational attainment at either margin.¹⁷ Overall, the effect of FDI deregulation on the gender education gap only operates at the sensitive margin of high school education.

lights. The nightlight strength variable is preferred since data on prefecture-level GDP per capita only cover urban areas and have many missing observations.

¹⁶In China, students attend three years middle school before high school, and three years high school before college. Therefore, we use $16 - 3 = 13$ as the key exposure age for middle school and $16 + 3 = 19$ as the key exposure age for college.

¹⁷Two factors may explain the insignificant results for college attendance. First, China’s gross tertiary enrollment rate stood at only 18.55% in 2005, substantially lower than the approximately 80.63% observed in the United States during the same period (World Bank, 2005). This educational structure suggests that FDI deregulation effects may be concentrated among cohorts with sub-tertiary educational attainment, given the limited penetration of higher education in China’s labor market during this period. Second, the prefecture of observed hukou registration may not correspond to the location where individuals actually experienced FDI shocks when making college attendance decisions. Chinese students commonly transfer their hukou registration to their university’s location upon enrollment, so identifying FDI exposure based on individuals’ prefecture of hukou registration may introduce measurement error.

Validity of the Shift-Share Design: Shock Exogeneity

Our empirical strategy exploits variations in industrial composition across regions and FDI deregulation shocks at the industry level. This empirical strategy belongs to a class of “shift-share design,” as popularized by [Bartik \(1991\)](#). The shift-share design either relies on the assumption of share exogeneity or that of shock exogeneity. Share exogeneity is the appropriate assumption if the variation mainly comes from a few specific industries ([Goldsmith-Pinkham, Sorkin, and Swift, 2020](#)).¹⁸ We use deregulation shocks for 436 industries (CIC 4-digit), which are even more than the 328 regions. We rely on the assumption of “shock exogeneity.”¹⁹

Based on the assumption of “shock exogeneity,” [Adão, Kolesár, and Morales \(2019\)](#) show that cross-region similarity in industrial composition results in a spatial correlation of standard errors and causes an over-rejection of the null hypothesis. The suggested inference procedure of [Adão, Kolesár, and Morales \(2019\)](#) does not apply when the data contain more industries than regions, as is our case. To solve the same issue, [Borusyak, Hull, and Jaravel \(2022\)](#) offer an alternative procedure that can be applied when there are more industries than regions.²⁰ Furthermore, [Borusyak, Hull, and Jaravel \(2022\)](#) show that the region-level

¹⁸We have also verified the latent validity on share exogeneity. On one hand, by computing Rotemberg weights across 4-digit CIC industries, we find that the five industries with the largest positive contributions account for a modest share among the total positive Rotemberg weight. On the other hand, we employ a leave-one-sector-out strategy confirming that no single industry appears to be overly influential in driving our results. Detailed discussions are provided in the Online Appendix A.1.

¹⁹We thank the editor and an anonymous referee for their valuable comments on the validity of the shift-share design.

²⁰We further probe the robustness of our statistical inference by re-estimating the baseline model with several alternative standard error clustering schemes for the shift-share design. First, we broaden the geographical clustering from the prefecture to the province level, a broader geographical unit than the prefecture. Second, following the approach of [Campante, Chor, and Li \(2023\)](#) and [Ma, Pan, and Xu \(2025\)](#) for constructing share-similarity which measures the congruence between a prefecture’s initial employment structure and that of its provincial capital, we cluster standard errors by these employment-similarity groups and, separately, by employment-similarity groups defined outside the original province. Finally, we implement two-way clustering: (i) at the province and employment-similarity group levels, and (ii) at the province and outside-of-province employment-similarity group levels. As reported in Online Appendix Table B4, our estimates remain statistically significant and consistent with baseline

regression has an equivalent representation at the industry level. Following [Borusyak, Hull, and Jaravel \(2022\)](#), we carry out robustness tests at the industry level.

We first examine the plausibility of “shock exogeneity.” At the 4-digit industry level, we regress the indicator of FDI deregulation in 2002 on industry characteristics in 2000. Similar to [Lu, Tao, and Zhu \(2017\)](#), the industry characteristics include the Herfindahl-Hirschman Index (HHI), new product share, export share, state-own-enterprise (SOE) share, average firm age (log scale), and the number of firms (log scale). Column (1) of Online Appendix Table B5 shows that the deregulation indicator correlates with three industry characteristics: SOE share, average firm age, and the number of firms. Industries with lower SOE shares, older firms, or smaller numbers of firms are more likely to be deregulated. In column (2), we control for the fixed effect of 2-digit industries and examine shock variations within each 2-digit industry. The coefficients on SOE share and average firm age decrease and lose statistical significance. “Shock exogeneity” is more likely to hold within 2-digit industries.²¹

We then transform the region-level data to the industry level and check the robustness of the results by adding industry-level controls. To facilitate the transformation, we aggregate our sample to the region level. The aggregate takes three steps, which we illustrate using the main outcome variable as an example. First, we calculate the region-level high school attendance rates of the treated (1986–1988) and untreated (1983–1985) cohorts for each gender. Second, we compute the gender gap (male over female) in high school attendance rates. Third, we take the cohort difference, i.e., the treated cohort relative to the untreated cohort, of the gender gap in high school attendance. These three steps deliver the cohort

results across all specifications. This indicates that our primary findings are robust to potential residual correlations arising from similarities in industrial structure across prefectures.

²¹Column (3) of Online Appendix Table B5 controls for the fixed effect of 3-digit industries, and shows results similar to column (2). We consider 164 fixed effects to be too numerous given that our data only include 436 4-digit industries. Therefore, we take results conditional on 3-digit industry fixed effects as suggestive only.

difference in the gender gap in high school attendance in 328 regions. We analogously aggregate other variables and follow [Borusyak, Hull, and Jaravel \(2022\)](#) to transform the region-level aggregate data to industry-level aggregate data.

Column (1) of Online Appendix Table B6 shows the estimate of the industry-level regression, using the transformed cohort difference of the gender gap in high school attendance as the outcome variable. We cluster the standard errors at 3-digit industries to allow for correlations of shocks inside 3-digit industries. Similar to our main results, FDI deregulation narrows the gender gap in high school attendance. Column (2) controls for the industry characteristics that we examined above. Column (3) further controls for the fixed effect of 2-digit industries, and column (4) controls for the fixed effect of 3-digit industries. In all columns, the effects of FDI deregulation remain negatively significant and qualitatively similar. Overall, our results are robust when we correct for spatial correlations of standard errors and purge out pre-determined industry confounders.²²

Validity of the Shift-Share Design: Randomization Inference

As demonstrated by [Borusyak and Hull \(2023\)](#), omitted variable bias (henceforth OVB) can emerge even when leveraging quasi-experimental exogenous variation if observations systematically receive different treatment intensities due to unobserved confounders. This is particularly relevant in settings where exposure differences are rooted in non-random factors, such as predetermined geographical location. Applied to our context, potential

²²Our shift-share measure of FDI deregulation features “incomplete shares,” where shares within a region do not sum up to a constant. We address the concern of incomplete shares in two steps. First, following [Borusyak, Hull, and Jaravel \(2022\)](#) and [Borusyak, Hull, and Jaravel \(2025\)](#), we control for the interaction terms between cohort dummies and the sum of shares in a region in Online Appendix Table B7. Second, we employ recentered FDI deregulation using the randomization inference approach ([Borusyak and Hull, 2023](#)), as presented in Online Appendix Table B9. Both sets of estimates remain quantitatively similar to our baseline results, suggesting that our shift-share design does not suffer from biases of incomplete shares.

sources of OVB may manifest through heterogeneous local industrial structures that generate differential labor demand by gender across regions (Kwon and Lee, 2024). Consequently, if the variation in FDI deregulation exposure is systematically correlated with these unobserved industrial structure factors, then our estimates would be subject to OVB and potentially inconsistent, even if the plausible exogeneity assumptions regarding the shift component of the instrument are met.

To bolster the validity of our shift-share design, we undertake additional robustness checks following the methodology proposed by Borusyak and Hull (2023). In particular, we construct both of expected and recentered instruments for FDI deregulation intensities to mitigate potential OVB stemming from the nonrandom exposure of localities to exogenous FDI deregulation shocks.

The process for generating recentered instruments of FDI deregulation shocks proceeds in three steps. First, we generate random draws for the industry-level FDI deregulation intensities by shuffling the deregulation indicator 3,000 times within each 3-digit industry. The resulting counterfactual values are then averaged to yield the expected FDI deregulation intensities. This expected instrument is designed to capture potential nonrandom exposure stemming from unobserved industrial characteristics across regions. Second, we aggregate the industry-level expected instrument to the regional level using a shift-share procedure analogous to that described in Eq. (1). Third, we construct the recentered instrument by subtracting the industry-level expected instrument from the actual FDI deregulation intensities. These recentered values are then aggregated via the shift-share design to yield region-level recentered instruments.

We first examine whether our recentered instrument effectively purges OVB. We conduct

a validity test for the prefecture-level measure of recentered FDI deregulation. Following [Borusyak and Hull \(2023\)](#), we employ a set of geographical factors as proxies for potential sources of predetermined nonrandom exposure to FDI deregulation shocks. As shown in Online Appendix Table B8, the recentered instrument exhibits no statistically significant correlation with either the expected instrument or an extensive set of initial prefecture-level characteristics.

We then employ two methodological approaches recommended by [Borusyak and Hull \(2023\)](#) to address potential endogeneity concerns: (1) utilizing recentered FDI deregulation intensity in place of the original measure, and (2) directly controlling for the expected instrument based on the original shock. Column (1) of Online Appendix Table B9 presents estimates obtained using the recentered instruments of FDI deregulation shocks, while column (2) reports results from the approach controlling for the expected instrument. The gender-differential effect of FDI deregulation on high school attendance remains positive and statistically significant across both specifications.

4 Mechanisms

In this section, we examine two channels through which FDI deregulation affects the gender gap in education. Numerous studies have shown that globalization affects people’s schooling choices by altering their opportunities in the labor market (e.g., [Atkin, 2016](#); [Li et al., 2019](#); [Dai, Huang, and Zhang, 2021](#)). For example, the establishment of foreign-invested enterprises creates employment opportunities for individuals aged 16 and above. This may shift the labor demand curve upward in a manner that differs by skill level. In particular, if deregulation primarily benefits industries that rely on low-skilled labor, this could incentivize

young individuals to forgo high school education in favor of an early entry into the workforce. Conversely, if deregulation stimulates a surge in demand for high-skilled workers, it may encourage individuals to invest in human capital to meet the requirements of emerging, creative job opportunities. Moreover, if this shift in labor demand does not affect both genders equally, it might lead to changes in the gender education gap.²³ We denote this channel as the **job-opportunity effect**.

FDI also diffuses gender-equal culture (Kodama, Javorcik, and Abe, 2018; Tang and Zhang, 2021). Tang and Zhang (2021) show that FDI transfers gender-equal social norms to Chinese workplaces by hiring more women and appointing more female managers. In addition to gender discrimination in the workplace, gender bias roots in the household. The patriarchal Chinese society embeds a preference for sons, which leads to intrahousehold resource allocation that favors sons (Qian, 2008; Zhang, 2019). FDI may diffuse more gender-equal culture to the locality, reduce son preference of Chinese parents, and induce gender-equal treatment in educational investment. We denote this channel as the **culture spillover effect**.

The culture spillover effect contributes to the gender education gap in a different way compared with the job opportunity effects. For example, FDI may expand high-skilled employment opportunities for women, thereby incentivizing higher educational attainment. By contrast, FDI may also convey gender-equal culture to suppress pre-existing son preference of Chinese parents, lead to equal educational investment in daughters and sons, and narrows the gender educational gap. The analysis in this section provides evidence that both channels may be responsible for the impact of FDI on the gender gap in education.

²³The female employment share is, on average, 0.48 in FDI firms, but only 0.39 in domestic firms, according to the 2004 Economic Census in China.

4.1 Job-opportunity Effect

FDI deregulation may induce gender-specific variations in the demand for skills, and shape the gender education gap. A specific feature of this FDI reform in 2002 is that deregulated industries are more female-skill-intensive. As shown in Online Appendix Table B10, female workers in reformed industries have 0.35 more years of schooling than female workers in unreformed industries (Panel C).

To capture gender-specific variations in skill demand, we follow [Li \(2021\)](#) to define female-oriented skills, and leverage on the importance scores of these skills for different occupations in the O*NET database. Based on the existing literature, [Li \(2021\)](#) suggest that women hold comparative advantages in language and social skills, while men excel in cognitive spatial skills, stamina, and muscular strength. In particular, female-oriented skills comprise eight skills spanning two domains. First, the language domain, including skills of oral comprehension, oral expression, written comprehension, and written expression. Second, the social domain, including skills of negotiation, persuasion, service orientation, and coordination.

We compute each occupation’s female-skill intensity, using the importance scores of female-oriented skills for different occupations in the the O*NET database. Online Appendix A.4 presents the details of calculation. Overall, occupations with high female-skill intensity are professional jobs, such as Public Relations Specialists, Sale Engineers, and Lawyers (Online Appendix Table A7a). For occupations most relevant to industries that experienced FDI deregulation, female-skill-intensive occupations mainly comprises white-collar jobs such as Clerks, Secretaries, Customer Service Representatives, and Retail Salespersons

(Online Appendix Table A7b).

We then construct industry-level female-skill intensity measures by computing weighted averages of occupational scores for each female-oriented skill, where weights are occupation-industry employment shares from the U.S. Occupational Employment Statistics Survey. This yields skill-specific female-skill intensity measures at the industry level. We apply principal component analysis to these measures to generate a composite index of relative female-skill intensity across industries. Using this index, we decompose the FDI shock described in Eq. (1) into a female-skill-intensive component and its reverse counterpart. Specifically, the female-skill-intensive FDI shock is constructed as follows:

$$FDI_r^F = \sum_k \frac{Emp_{kr0}}{Emp_{k0}} \cdot Treatment_k \cdot SkillInt_k^F, \quad (5)$$

where $SkillInt_k^F$ denotes the relative importance of female-oriented skills in industry k ,²⁴ and $Treatment_k$ is an indicator variable equal to one if industry k experienced FDI reform, and zero otherwise. The variable FDI_r^F represents the prefecture-level female-skill-intensive FDI shocks.²⁵ Analogously, we define the male-skill-intensive FDI shocks as follows:

$$FDI_r^M = \sum_k \frac{Emp_{kr0}}{Emp_{k0}} \cdot Treatment_k \cdot (1 - SkillInt_k^F), \quad (6)$$

where $(1 - SkillInt_k^F)$ measures the relative importance of male-oriented skills in industry k . In this way, we have decomposed our overall FDI shock into two gender-specific shocks:

²⁴The relative importance of female-oriented skills represents the degree to which an industry utilizes skills where females possess comparative advantage relative to males. For detailed construction methodology and definitions of the measure on relative female-oriented skills importance, see Online Appendix A.3.

²⁵Our measure of female-skill-intensive FDI shock is similar to the measures of overall female-oriented skill export intensity constructed by Li (2021), thereby capturing the most comprehensive information regarding gender-specific skills in FDI deregulation.

$$FDI_r = FDI_r^F + FDI_r^M.$$

We examine how female-skill-intensive FDI deregulation affects the ratio of high-skilled (or low-skilled) female to male employment, using the regression specification below:

$$S_{nrt} = \beta_{1n} FDI_r^F \cdot Post_t + \beta_{2n} FDI_r^M \cdot Post_t + \mathbf{Z}_{rt}' \mathbf{\Gamma} + \lambda_r + \lambda_{pt} + \varepsilon_{rt}, \quad (7)$$

where S_{nrt} is the ratios of high-skilled (or low-skilled) female to male in the non-agricultural employment for age group n in prefecture r and year t .²⁶ We employ a repeated cross-sectional design using prefecture-level data from the 1990, 2000, 2005, and 2010 censuses. The coefficient β_{1n} captures the effect of the female-skill-intensive FDI liberalization for age group n . In parallel, the coefficient β_{2n} estimates the effect of the maleskillintensive FDI shock.

Columns (1)–(4) of Table 4 show the estimates of Eq. (7) for high-skilled workers. Column (1) reports that a femaleskill-intensive FDI deregulation raises the highskill female-to-male employment ratio for individuals aged 16–60 by 8.3 percentage points. In comparison, the maleskill-intensive FDI shock lowers this ratio by approximately 4.3 percentage points, with both effects statistically significant. We further disaggregate the sample by age group in columns (2)–(4), corresponding to workers aged 16–30, 31–45, and 46–60, respectively. The estimated effects remain large in magnitude and statistically significant at the 1% level for the youngest cohort, aged 16–30 (column (2)). While we also find positive and significant estimates for workers aged 31–45 (column (3)), the point estimates are roughly half the size of those for the younger group. For the oldest cohort (column (4)), the estimates are

²⁶High-skilled workers are defined as those with at least a high school education, while low-skilled workers are those with education below the high school level.

small and not statistically distinguishable from zero. Taken together, these findings suggest that the employment effects of FDI deregulation for high-skilled women are concentrated primarily among younger workers.

By contrast, FDI deregulation does not influence the female-to-male employment ratios for low-skilled workers across any of the age groups examined (columns (5)–(8)). These results align with our theoretical hypothesis that easing FDI restrictions boosts employment opportunities for young skilled women, thereby increasing the expected returns to human capital investment and potentially incentivizing parents to invest in their daughters’ education to meet rising demand for educated female workers.

4.2 Culture-spillover Effect

Another important possibility is that FDI deregulation might affect the education gender gap through cultural diffusion, as documented by a limited but growing strand of literature (Kodama, Javorcik, and Abe, 2018; Tang and Zhang, 2021). Gender inequality has been deeply rooted in cultural norms (Alesina, Giuliano, and Nunn, 2013; Jayachandran, 2015). The job-opportunity effect, which we examined in the previous subsection, tends to depend on the industry particularities of the specific FDI episode. In comparison, the culture-spillover effect is generally at work when the FDI flows to gender-unequal destinations. The culture-spillover effect works via an attenuation of pre-existing gender bias, with improved female labor market conditions under FDI liberalization serving as the transmission channel. We hypothesize that, following FDI deregulation, genderunequal cultural norms will be attenuated. In this subsection, we empirically assess this prediction.

Our heterogeneity analysis (Table 2 in Section 3.1) highlights the pivotal role of gender-

equal cultural norms in shaping the effect of FDI on the gender education gap. However, one may doubt whether FDI deregulation indeed suppresses gender-unequal culture. In this section, we explore this issue with more evidence.

The male-biased birth sex ratio has been regarded as a revelation of gender-unequal values and beliefs (Jayachandran, 2015; Xue, 2024). In China, the birth sex ratio became highly imbalanced after the 1990s, when the one-child policy tightened and diagnostic ultrasound became widely available (Ebenstein, 2010; Chen, Li, and Meng, 2013). The birth sex ratio is particularly informative on son preference when the expected fertility level is low (Jayachandran, 2017). As China’s birth control policy mandates low fertility, the birth sex ratio informs son preference in China.

We examine whether the FDI shock affects the regional sex ratios at birth. Specifically, we calculate prefecture-level sex ratios before and after FDI deregulation based on county-level statistics for the China Population Censuses 2000 and 2010. The sex ratio is defined by the number of males divided by the number of females. For the two census years (2000 and 2010),²⁷ we use prefecture-level sex ratios for people at age 0 and for people in five-year age groups: 1–4, 5–9, and 10–14. We do not use sex ratios for older age groups because people older than 15 may migrate across prefectures.

In the 2000 census, people at age 0 were born in 2000, before FDI deregulation in 2002. In the 2010 census, people at age 0 were born in 2010, after FDI deregulation. Therefore, people at age 0 in the 2000 and 2010 census serve as the pre- and post-treatment groups, respectively. Similarly, age groups 1–4 in the 2000 (2010) census, born in 1996–1999 (2006–2009), can also serve as the pre- and post-treatment groups. To examine the effect of FDI

²⁷The sex ratio remained highly imbalanced and declined slightly over 2000 and 2010. The average sex ratio of the 1–4 age group declined by approximately two percentage points from 1.184 to 1.163 between 2000 and 2010.

deregulation on son preference, we estimate the region-group-year specification analogous to Eq. (7), using the sex ratio at age 0 or the sex ratio in age groups 1–4 as the dependent variable. For older age groups (5–9 and 10–14), people from both the 2000 and 2010 censuses do not have sufficient exposure to FDI deregulation. Therefore, estimates of sex ratios of older age groups (5–9 and 10–14) offer placebo tests.

Panel A of Table 5 reports estimates using the prefecture-level sex ratios of certain age groups as the dependent variable. Columns (1)–(4) present the results for sex ratios at age (group) 0, 1–4, 5–9, 10–14, respectively. Columns (1) and (2) show that FDI deregulation reduces the male-biased sex ratios for age 0 and age groups 1–4, which contain people born in 1996–2000 ($Post = 0$) and in 2006–2010 ($Post = 1$). We also find that FDI deregulation does not affect the sex ratios for older age groups whose parents have not been fully exposed to FDI deregulation (columns (3)–(4)). These placebo tests suggest that FDI deregulation does not capture pre-determined trends of sex ratio imbalance.

The literature takes the birth sex ratio as a revelation of gender-biased culture. However, numerous studies have also shown that the birth sex ratio can be affected by the relative costs and benefits of raising sons versus daughters (e.g., [Qian, 2008](#); [Guo et al., 2022](#); [He et al., 2025](#)). It is possible that the effect of FDI on the birth sex ratio may still reflect the gender-differential effects of FDI on future earnings and job opportunities. If this job-opportunity channel dominates, we would expect the suppressing effect of FDI on the sex ratio imbalance to exist in both regions with weaker and stronger gender-unequal culture norms. On the other hand, if the culture effect is the driving force, we would expect a more salient effect in areas with a stronger gender-unequal culture. In Panels B and C of Table 5, we split the sample by genealogy density, a measure of preexisting gender-biased culture.

Consistent with the culture-spillover effect, the effects of FDI deregulation on sex ratios are driven by areas with above-median genealogy density. The effects are absent in areas with below-median genealogy density. Overall, the results on sex ratios provide direct evidence that FDI deregulation reduces son preference in China.

5 Longer-run Effects on Female Career

Increasing evidence shows that empowering women could have long-run consequences, such as benefiting long-run economic growth by promoting education and children’s performance (Thomas, 1990; Qian, 2008; Duflo, 2012). Our findings on FDI-induced education-work choice at a key exposure age reflect a short-run trade-off a few years following the FDI shock. Females who choose to attend school after the FDI shock will complete their education and enter the labor market. Would these different educational choices eventually translate into better labor market performance for females than for males in the long run?

In this section, we examine the gender-differential consequences of the FDI shock on labor market outcomes in 2010, eight years after the reform. Online Appendix Table B11 summarizes the labor market outcomes for people born in 1983–1988 in the 2010 census. Our sample individuals reached 24 years old on average in 2010. As most of our sample have passed school age, most are employed: 86% of males and 75% of females have jobs. Similarly, the labor participation rate (LFP) is 90% for males and 80% for females. Consistent with the literature (Chen and Ge, 2018), the female LFP rate is particularly high in China. Conditional on employment, males are 2.6 percentage points less likely to do white-collar jobs than females are.

Columns (1)–(3) of Panel A in Table 6 show the estimates from the interaction specifi-

cation given by Eq. (2), along with separate estimates for each gender, using labor market outcomes in 2010 as the dependent variables. We find that eight years after the FDI relaxation in 2002, FDI deregulation increases the likelihood of being employed for both males and females, but the effects are statistically insignificant for both genders (columns (1) and (2) of Panel A).²⁸ We do not detect a gender-differential effect of FDI on the likelihood of being employed in 2010 (column (3) of Panel A).

We then examine the longer-run effect of FDI on holding a white-collar job in columns (4)–(6) of Panel A. We find a positive effect of FDI deregulation on the likelihood of holding a white-collar job for females but not for males (columns (4) and (5) of Panel A). By column (6), the gender-differential effect is statistically significant and economically meaningful. A median increase in FDI deregulation raises the likelihood of females doing a white-collar job relative to males by $0.031 \times 0.140 \times 100 = 0.43$ percentage points, which explains 16.5% of the 2.6 percentage-point advantage of females. The results suggest that the FDI-induced educational advantage of females translates into esteemed occupations in the longer run.²⁹

In Panel B and C of Table 6, we split the sample by genealogy density and reestimate the Panel A specifications. Column (5) of Panel B shows that the positive impact of FDI deregulation on white-collar employment is driven entirely by regions with abovemedian genealogy density, consistent with a culturespillover mechanism.

²⁸Results are similar if we use labor force participation as the outcome variable instead of employment status.

²⁹This finding aligns with the descriptive evidence in Online Appendix Table A7, which indicates that FDI deregulation expands the employment opportunity for white-collar occupations that intensively use female-oriented skills.

6 Conclusion

Empowering women could benefit the long-run growth of the economy (Thomas, 1990; Qian, 2008; Duflo, 2012). How globalization empowers women has attracted great attention from both academia and the policy arena. While most studies focus on how international trade affects the gender gap, our research focuses on the effect of FDI, another important phenomenon of globalization.

By exploiting regional variations in a quasi-natural FDI shock due to the deregulation of FDI entry in China, we identify a causal effect of FDI in narrowing the education gender gap. This reduction operates through two distinct channels. First, female-skill-intensive FDI entries expand the relative demand for skilled young female workers, which makes females relatively more likely to continue high school education. In addition, we also find evidence that FDI entry affects the gender education gap by diffusing a gender-equal culture. FDI deregulation has a significant effect on the gender gap in education in regions with a strong Confucian culture, and regions more exposed to FDI shocks tend to have an alleviation of son preferences, especially regions with entrenched gender bias. Moreover, FDI deregulation also delivers persistent benefits for female careers, manifested in higher white-collar employment than their male counterparts eight years after the reform.

These insights underscore FDI's distinctive capacity to promote inclusive human capital development by simultaneously altering economic incentives and cultural attitudes. Our findings suggest that policies encouraging FDI can be instrumental in advancing gender equality and empowering women through education, especially in countries with entrenched gender biases.

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7 Tables

Table 1: Summary statistics of individual variables

Sample	Full		Male		Female	
	Mean (1)	S.D. (2)	Mean (3)	S.D. (4)	Mean (5)	S.D. (6)
Education						
Middle school	0.911	0.284	0.924	0.265	0.899	0.302
High school	0.389	0.487	0.406	0.491	0.372	0.483
College	0.18	0.384	0.18	0.384	0.18	0.384
Demographics						
Age	22.675	2.875	22.676	2.887	22.675	2.864
Han	0.897	0.304	0.896	0.305	0.897	0.303
Observations	698,407		343,087		355,320	

Note: The data source is the 2005 and 2010 waves of the China population (mini-)census. We include people born in 1983–1988 who were 14–19 years old at the relaxation of the FDI regulations in 2002. Differences in average college attendance rates between the full sample and subsamples by gender are evident only at the fourth decimal place.

Table 2: Baseline results and heterogeneity

Dependent variable	High-school attendance				
Sample	Hukou status			Genealogy density	
	Full	Rural	Urban	Above-median	Below-median
	(1)	(2)	(3)	(4)	(5)
$FDI \times Age16 \times Female$	0.056*** (0.017)	0.050** (0.020)	0.061** (0.027)	0.100*** (0.021)	-0.015 (0.033)
$FDI \times Age16$	-0.038** (0.015)	-0.023 (0.018)	-0.070*** (0.020)	-0.055** (0.021)	-0.008 (0.027)
Contemporaneous policy controls	Yes				
Initial conditions # cohorts	Yes				
Fixed effects	Prefecture, Province-by-cohort				
Observations	698,407	544,445	153,962	301,280	299,055
R-squared	0.27	0.09	0.08	0.24	0.26

Note: The sample includes people born in 1983–1988 in the 2005 and 2010 waves of the China population (mini-)census. They were 14–19 years old in 2002, when the FDI regulations were relaxed. Column (1) shows the result for the full sample. Hukou status is defined as the classification assigned to an individuals official household registration. Column (2) presents the analysis for individuals with rural hukou registrations, whereas Column (3) reports results for those with urban hukou registrations. Genealogy density measures the strength of Confucian culture. Column (4) uses the sample of prefectures with above-median density of genealogies, and Column (5) uses the sample of prefectures with below-median density of genealogies. All regressions includes control variables including ethnicity, urban status, prefecture fixed effect and province-by-cohort fixed effect. In addition, all specifications interact cohort dummies with predetermined prefecture characteristics, and control for contemporaneous WTO accession policies. The predetermined prefecture characteristics include average years of schooling, the share of agricultural employment, and the share of manufacturing employment among individuals aged 1660 as recorded in the 1990 China population census. Contemporaneous confounders are addressed by interacting cohort dummies with both the initial import tariff from 2001 and the column 2 tariff, respectively. All control variables are interacted with the female dummy to allow for gender-specific effects. Standard errors in parentheses are clustered by prefecture. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Robustness

Dependent variable	High-school attendance					
	Vertical linkages (1)	Neighborhood regions (2)	Inverse distance (3)	Excluding migrants (4)	Economic development (5) (6)	
$FDI \times Age16 \times Female$	0.060*** (0.017)	0.057*** (0.017)	0.056*** (0.018)	0.041** (0.019)	0.057*** (0.018)	0.057*** (0.018)
$FDI \times Age16$	-0.039*** (0.015)	-0.038** (0.015)	-0.040** (0.015)	-0.029* (0.015)	-0.039*** (0.015)	-0.040*** (0.015)
$FDI_{up} \times Age16 \times Female$	-0.016 (0.027)					
$FDI_{up} \times Age16$	-0.013 (0.023)					
$FDI_{down} \times Age16 \times Female$	-0.021 (0.014)					
$FDI_{down} \times Age16$	0.017 (0.014)					
$FDI_{neighbor} \times Age16 \times Female$		0.031 (0.044)				
$FDI_{neighbor} \times Age16$		-0.009 (0.032)				
$FDI_{other} \times Age16 \times Female$			0.296 (0.520)			
$FDI_{other} \times Age16$			-0.174 (0.423)			
GDP per capita # cohort					Yes	
Nightlight strength # cohort						Yes
Controls				Yes		
Fixed effects				Prefecture, Province-by-cohort		
Observations	698,407	698,407	698,407	576,850	602,011	698,407
R-squared	0.27	0.27	0.27	0.28	0.25	0.27

Note: The sample includes people born in 1983–1988 in the 2005 and 2010 waves of the China population (mini-) census. Control variables are the same as those in Table 2. All control variables are interacted with the female dummy to allow for gender-specific effects. Standard errors in parentheses are clustered by prefecture. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: The ratio of female to male employment by skill and age

Dependent variable	The ratio of female to male employment							
	High skill				Low skill			
	16–60 (1)	16–30 (2)	31–45 (3)	46–60 (4)	16–60 (5)	16–30 (6)	31–45 (7)	46–60 (8)
$FDI^F \times Post$	0.083*** (0.027)	0.167*** (0.054)	0.073** (0.036)	-0.009 (0.030)	-0.033 (0.047)	-0.072 (0.083)	0.052 (0.057)	-0.024 (0.049)
$FDI^M \times Post$	-0.043*** (0.016)	-0.102*** (0.032)	-0.035 (0.022)	0.007 (0.018)	0.024 (0.025)	0.058 (0.047)	-0.015 (0.036)	0.001 (0.026)
Controls	Yes							
Fixed effects	Prefecture, Province-by-wave							
Observations	1,310	1,310	1,310	1,310	1,310	1,310	1,310	1,310
R-squared	0.82	0.62	0.83	0.72	0.77	0.70	0.75	0.72

Note: This table reports the estimated effects of female-skill-intensive and male-skill-intensive FDI shocks on the relative employment of females compared to males in non-agricultural sectors. The dependent variables are the ratios of female to male employment for people aged 16–60, 16–30, 31–45 or 46–60 by skill level. Columns (1)–(4) restrict the sample to high-skilled workers, defined as those with a high school education or above, while columns (5)–(8) restrict the sample to low-skilled workers, defined as those with less than a high school education. Control variables include the share of Han, the share of males, tariff and uncertainty shocks, and interaction between wave dummies and a set of pre-determined prefecture characteristics. The tariff and uncertainty shocks include interactions of the initial import tariff and column 2 tariff levels with the Post dummy. The predetermined prefecture characteristics are the same as those in Table 2. The sample includes prefecture-level data calculated from the 1990, 2000, 2005 and 2010 waves of the China population censuses. Standard errors in parentheses are clustered by prefecture. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: The effects of FDI deregulation on sex ratio imbalance

Dependent variable	Sex ratio			
Age group	0 (1)	1–4 (2)	5–9 (3)	10–14 (4)
Panel A. Full sample				
$FDI \times Post$	-0.066** (0.030)	-0.069*** (0.024)	-0.021 (0.019)	-0.029 (0.031)
Observations	652	652	652	652
R-squared	0.89	0.92	0.95	0.86
Panel B. Above-median genealogy density				
$FDI \times Post$	-0.164*** (0.043)	-0.111*** (0.034)	-0.016 (0.029)	-0.004 (0.042)
Observations	250	250	250	250
R-squared	0.89	0.92	0.95	0.88
Panel C. Below-median genealogy density				
$FDI \times Post$	0.006 (0.060)	0.066 (0.088)	0.002 (0.053)	-0.060 (0.071)
Observations	244	244	244	244
R-squared	0.89	0.93	0.95	0.86
Controls	Yes			
Fixed effects	Prefecture, Province-by-wave			

Note: The dependent variable is the prefecture-level sex ratio for a certain age group. The sex ratio is defined by the number of males divided by the number of females. We calculate prefecture-level sex ratios from the county-level statistics for the China Population Censuses of 2000 and 2010. Columns (1)–(4) present the results for sex ratios at age (group) 0, 1–4, 5–9, and 10–14, respectively. Panel A shows results for the full sample, Panel B for prefectures with above-median genealogy density, and Panel C for prefectures with below-median genealogy density. The number of observations in Panel A is larger than the sum of those in Panels B and C because genealogy density is not available for 79 matched prefectures. Control variables are the same as those in Table 4. Standard errors in parentheses are clustered by prefecture. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Longer-run effects on employment and occupation

Dependent variable	Employment status in 2010			White-collar job in 2010		
	Male (1)	Female (2)	Full (3)	Male (4)	Female (5)	Full (6)
Panel A. Full sample						
$FDI \times Age16$	0.027 (0.020)	0.024 (0.019)	0.027 (0.020)	0.002 (0.013)	0.033** (0.015)	0.002 (0.013)
$FDI \times Age16 \times Female$			-0.004 (0.018)			0.031* (0.017)
Observations	244,651	251,028	495,679	210,752	188,131	398,883
R-squared	0.14	0.07	0.11	0.13	0.22	0.18
Panel B. Above-median genealogy density						
$FDI \times Age16$	-0.003 (0.027)	0.001 (0.026)	-0.003 (0.027)	0.007 (0.018)	0.052*** (0.019)	0.007 (0.018)
$FDI \times Age16 \times Female$			0.004 (0.020)			0.045 (0.027)
Observations	105,650	107,692	213,342	91,467	82,432	173,899
R-squared	0.13	0.06	0.10	0.12	0.21	0.17
Panel C. Below-median genealogy density						
$FDI \times Age16$	-0.015 (0.037)	0.046 (0.030)	-0.015 (0.037)	-0.000 (0.020)	-0.022 (0.019)	-0.000 (0.020)
$FDI \times Age16 \times Female$			0.061 (0.040)			-0.022 (0.022)
Observations	106,613	108,814	215,427	93,112	82,344	175,456
R-squared	0.13	0.07	0.11	0.14	0.23	0.19
Controls	Yes					
Fixed effects	Prefecture, Province-by-cohort					

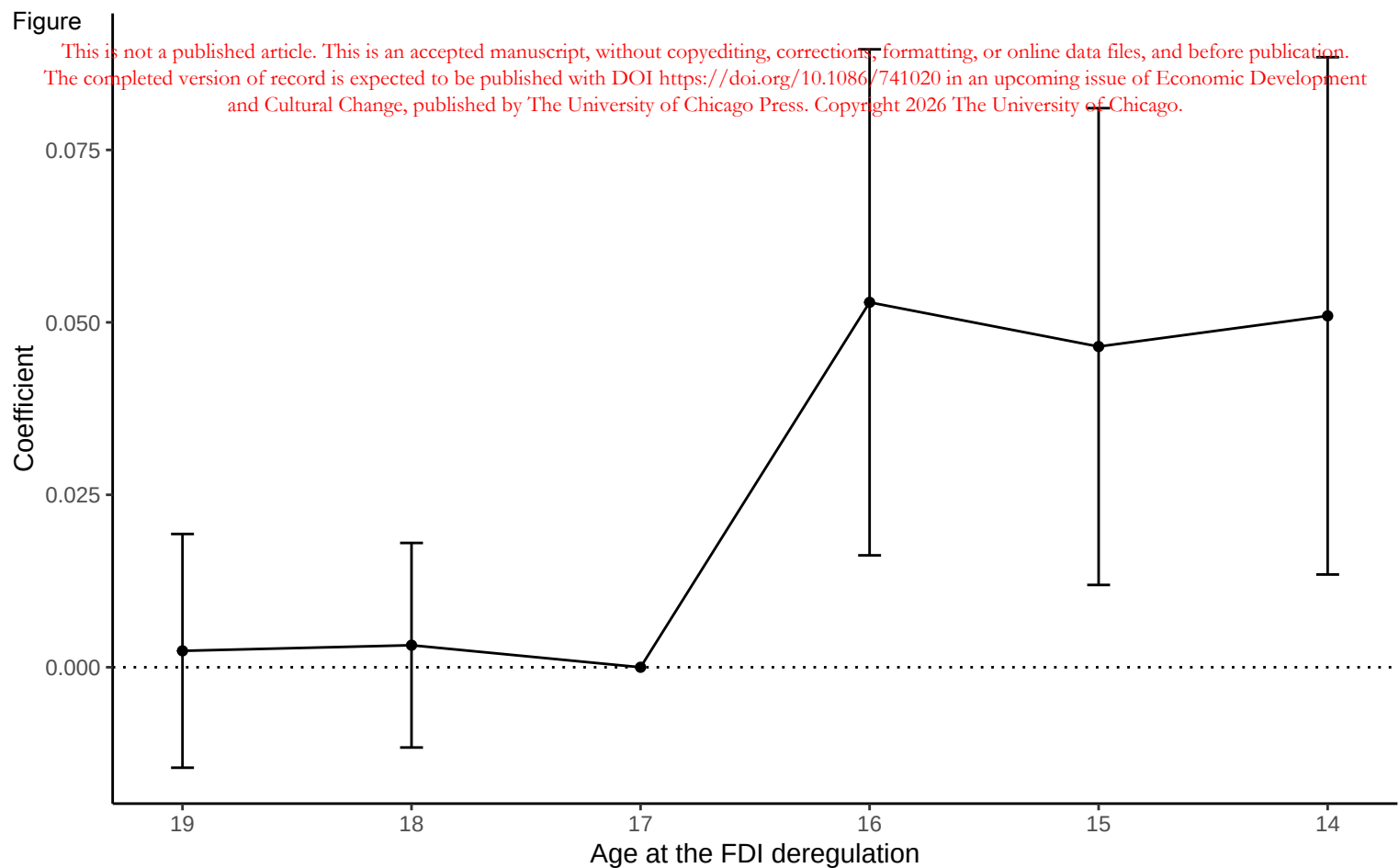
Note: The sample includes people born in 1983–1988 in the 2010 wave of the China population census. They were 14–19 years old in 2002, when the FDI regulations were relaxed. Control variables are the same as those in Table 2. All control variables are interacted with the female dummy to allow for gender-specific effects. Standard errors in parentheses are clustered by prefecture. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

8 Figure Legends

Figure 1. Event-study estimates

Note: This figure plots event-study estimates of β_{2a} from the specification in Eq. (3), with indicator of high school attendance as the dependent variable. The x-axis measures people’s age relative to the timing of FDI deregulation. Vertical bars represent confidence intervals at the 95% level. Data are from the 2005 and 2010 waves of the China population (mini-)census includes sample born during 1983–1988. Control variables are the same as those in Table 2. All control variables are interacted with the female dummy to allow for gender-specific effects. Standard errors in parentheses are clustered by prefecture.

Alt text: This figure presents coefficient estimates with 95% confidence intervals for the effect of FDI deregulation on the gender gap in high school attendance, plotted by age at the time of deregulation. The estimates show no statistically significant effects for individuals aged 18–19, cohorts who had already surpassed the typical age of high school enrollment decisions. In contrast, the estimates indicate a statistically significant narrowing of the gender education gap for individuals aged 16 or younger—cohorts who had not yet reached the critical decision age at the time of the policy change.



Foreign Direct Investment and Gender Education Gap

Appendix for Online Publication

Rufei Guo Haiji Lin Xin Wang Chaoqun Zhan*

Appendix A Data Details and Additional Empirical Analysis

A.1 Discussion on Share Exogeneity

To further substantiate the validity of our shift-share design for FDI deregulation, we investigate the exogeneity of its initial share components ([Goldsmith-Pinkham, Sorkin, and Swift, 2020](#)). Specifically, we conduct two tests to examine the exogeneity of industry employment shares across prefectures. First, we compute Rotemberg weights across 4-digit Chinese Industrial Classification (CIC) industries to assess the measure’s sensitivity to potential violations of the share exogeneity assumption. Second, we implement a leave-one-sector-out procedure to ensure that our baseline results are not driven by any single industry.

We first compute Rotemberg weights following the methodology proposed by [Goldsmith-Pinkham, Sorkin, and Swift \(2020\)](#). The estimation employs a cohort-difference specification that compares the gender gap in high school attendance between treated and untreated cohorts, consistent with our shock exogeneity analysis in Section 3.2—and includes province fixed effects alongside controls for potential confounders.¹ Industry-specific Rotemberg weights quantify the degree of sensitivity of shift-share estimates to violations of the share exogeneity assumption, with high-weight industries contributing most to potential bias.

*Rufei Guo, Center for Economic Development Research, Center for Health Economics and Management, Wuhan University. Haiji Lin, Center for Health Economics and Management, Economics and Management School, Wuhan University. Xin Wang, Institute of New Structural Economics, Peking University. Chaoqun Zhan (Corresponding author), Department of Accountancy, Economics and Finance, Hong Kong Baptist University, email: cqzhan@hkbu.edu.hk.

¹Controls include contemporaneous tariff reductions and trade policy uncertainty, cohort-differenced Han ethnicity share and urban hukou share, as well as predetermined agricultural employment share, manufacturing employment share, and average schooling years.

Table A1: Share exogeneity: Rotemberg weights

CIC 4-digit	Industry description	Weight
2665	Chemicals for information technology manufacturing	0.141
3662	Manufacturing of special-purpose equipment for the electronics industry	0.095
3792	Manufacturing of metal signs and facilities for traffic management	0.070
2613	Inorganic salt manufacturing	0.061
2190	Furniture manufacturing	0.059
Top five Rotemberg weights as a share of positive weight:		25.63%

Note: The table presents the five CIC 4-digit industries exhibiting the highest Rotemberg weights for FDI_r , alongside the proportion of the aggregate positive Rotemberg weight accounted for by these leading industries.

Table A1 reveals that the top five 4-digit CIC industries collectively account for approximately 25% of the total positive weight in our estimator. This level of concentration is considerably lower than documented by Goldsmith-Pinkham, Sorkin, and Swift (2020) in other prominent shift-share applications, where the top five industries frequently represented nearly 50% of the positive weight (e.g., Bartik, 1991; Card, 2009; Autor, Dorn, and Hanson, 2019).

Next, we employ a leave-one-sector-out procedure to assess whether our shift-share estimator is primarily driven by specific industries. This involves iteratively re-constructing the shift-share measure of FDI deregulation by omitting one 4-digit CIC industry at a time. We then re-estimate our baseline specification (Column (1), Table 2) with each of these modified measures. In total, the baseline model was re-estimated 426 times, corresponding to the number of 4-digit CIC industries. As presented in Table A2, we focus on the resulting minimum and maximum coefficients for the gender differential interaction term ($FDI \times Age16 \times Female$). Both the maximum and minimum estimates are comparable to our baseline coefficient, and the term remains statistically significant across these iterations. This indicates that our baseline findings are unlikely to be driven by any particular industry.

Our identification strategy prioritizes the exogeneity of the national industry-level shocks (FDI deregulation) over the exogeneity of the initial local industry employment shares. This emphasis is because local employment shares may be endogenous to various unobserved local factors, potentially confounding the estimated gender differential effects of FDI deregulation if the shares were to violate the exclusion restriction (Goldsmith-Pinkham, Sorkin, and Swift, 2020; Borusyak, Hull, and Jaravel, 2025). Therefore, while we present tests related to share exogeneity, we consider them supplementary, with our primary identification assumption

Table A2: Share exogeneity: Leave-One-Sector-Out

Dependent variable	High-school attendance	
	Min coef (1)	Max coef (2)
$FDI \times Age16 \times Female$	0.044*** (0.017)	0.056*** (0.016)
$FDI \times Age16$	-0.029* (0.016)	-0.036*** (0.013)
Controls	Yes	
Fixed effects	Prefecture, Province-by-cohort	
Observations	698,407	698,407
R-squared	0.27	0.27

Note: The table presents the minimum and maximum estimated coefficients for the gender differential effect of FDI deregulation, obtained by iteratively excluding one 4-digit CIC industry at a time. Contemporaneous confounders are subjected to a similar leave-one-sector-out procedure to maintain consistency. Other control variables are the same as those in Table 2. All control variables are interacted with the female dummy to allow for gender-specific effects. Standard errors in parentheses are clustered by prefecture. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

resting on the exogeneity of the shocks.

A.2 Robustness Checks: Heterogeneity in Gender-unequal Cultures

While the subsample analysis on gender-unequal culture in Table 2 is compelling, it is worth considering that genealogy density may capture factors beyond gender-unequal culture. For example, if regions with high genealogy densities already had a larger gender education gap before the FDI deregulation, the results in Table 2 may capture a greater decline in the gender education gap in regions where the initial gap was larger.

To address this concern, we utilize the gender gap in schooling years for the 1970–1980 cohorts as a measure of the pre-existing gender education gap. We find that genealogy density does not correlate with pre-existing gender gap (correlation coefficient = -0.048). Furthermore, it is important to note that high genealogy density represents the strength of Confucian culture in general, which serves as an informal institution to safeguard the development of China’s private sector (Zhang, 2020). We find that genealogy density positively correlates with nightlight strength in 2000 (correlation coefficient = 0.332) and the employment share of the private enterprises in 2000 (correlation coefficient = 0.103). To mitigate the influence

of potential confounding factors, we estimate Eq. (1):

$$\begin{aligned}
 Y_{irc} = & \theta_1 FDI_r \cdot Age16_c + \theta_2 FDI_r \cdot Age16_c \cdot Female_i + \\
 & \theta_3 FDI_r \cdot Age16_c \cdot Genealogy_r + \theta_4 FDI_r \cdot Age16_c \cdot Female_i \cdot Genealogy_r + \\
 & \theta_5 FDI_r \cdot Age16_c \cdot Confounder_r + \theta_6 FDI_r \cdot Age16_c \cdot Female_i \cdot Confounder_r + \\
 & \mathbf{X}'_{irc} \Gamma + \lambda_{gr} + \lambda_{gpc} + \varepsilon_i,
 \end{aligned} \tag{1}$$

where *Genealogy_r* is an indicator of high genealogy density, and *Confounder_r* represents a potential confounder such as pre-existing gender education gap, nightlight strength, or private sector share. We expect $\theta_4 > 0$, indicating that high genealogy density, conditional on the specific confounder, still magnifies the positive effect of FDI deregulation on female high-school attendance. Table A3 shows the estimates of Eq. (1). Column (1) does not include any confounders. Column (2) controls for pre-existing gender education gap, column (3) for nightlight strength, and column (4) for the share of private enterprises. The estimates of θ_4 is positive, statistically significant, and stable across all four columns.

Because the regressions in Table A3 includes quadruple interaction terms, we list all the main and sub-interaction terms of these regressions in Tables A4 and A5 to ensure adequate inclusion of controls.

Table A3: Potential confounders for genealogy density

Dependent variable	High-school attendance			
	None	Preexisting	Nightlight	Share of private
Confounder	(1)	gender gap	strength	enterprises
		(2)	(3)	(4)
$FDI \times Age16$	0.000 (0.025)	0.000 (0.027)	0.000 (0.027)	0.012 (0.026)
$FDI \times Age16 \times Female$	-0.024 (0.027)	-0.027 (0.030)	-0.010 (0.031)	-0.025 (0.029)
$FDI \times Age16 \times Genealogy$	-0.059* (0.034)	-0.068** (0.032)	-0.065 (0.041)	-0.063* (0.034)
$FDI \times Age16 \times Female \times Genealogy$	0.130*** (0.035)	0.132*** (0.039)	0.144*** (0.039)	0.134*** (0.035)
$FDI \times Age16 \times Confounder$		0.040 (0.048)	0.002 (0.023)	-0.278 (0.327)
$FDI \times Age16 \times Female \times Confounder$		-0.014 (0.047)	-0.030 (0.023)	-0.161 (0.267)
Controls		Yes		
Fixed effects		Prefecture, Province-by-cohort		
Observations	600,335	600,335	600,335	600,335
R-squared	0.25	0.25	0.25	0.25

Note: The sample includes people born in 1983–1988 in the 2005 and 2010 waves of the China population (mini-)census. They were 14–19 years old in 2002, when the FDI regulations were relaxed. $Genealogy_r$ is an indicator variable on above-median intensity of genealogies. For $Confounder_r$, Column (2) includes the gender gap in schooling years for the 1970–1980 cohorts, column (3) includes nightlight strength in 2000, and column (4) includes the employment share of private enterprises in the manufacturing sector in 2000. Control variables are listed in Tables A4 and A5. Standard errors in parentheses are clustered by prefecture. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A4 presents all right-hand-side variables included in column (1) of Table A3. At the start, we list all fixed effects and trends by predetermined conditions. We then introduce the remaining variables hierarchically: first level terms, then two-way interactions, followed by triple interactions, and finally quadruple interaction terms. For each term, we provide a brief description and state whether it is absorbed by any fixed effect. Similarly, Table A5 describes all right-hand-side variables included in columns (2)–(4) of Table A3.

Table A4: Detailed terms in Column (1) of Table A3

Specific term	Description
Fixed effects and trends	
<i>Prefecture-Gender Fixed Effects</i>	Capturing time-invariant characteristics that vary by prefecture and gender.
<i>Province-Cohort-Gender Fixed Effects</i>	Capturing province-by-gender-specific flexible trends.
<i>RegionChar</i> $\times$ <i>Cohorts</i>	Controlling for flexible trends in initial regional characteristics. ^a
<i>Genealogy</i> $\times$ <i>Cohorts</i>	Controlling for flexible trends in initial genealogy density across prefectures.
Level terms	
<i>FDI</i>	Prefecture-level FDI deregulation, absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Tariff</i>	Prefecture-level import tariff reduction, absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>PNTR</i>	Prefecture-level Column 2 tariff rate, absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Age16</i>	Absorbed by <i>Province-Cohort-Gender Fixed Effects</i>
<i>Female</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Genealogy</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>RegionChar</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>IndivChar</i>	Individual characteristics, i.e. indicators of Han ethnicity and Hukou status.
Two-way interaction terms	
<i>FDI</i> $\times$ <i>Age16</i>	The interaction terms of FDI deregulation with age indicator.
<i>Tariff</i> $\times$ <i>Age16</i>	The interaction terms of import tariff reduction with age indicator.
<i>PNTR</i> $\times$ <i>Age16</i>	The interaction terms of PNTR intensity with age indicator.
<i>FDI</i> $\times$ <i>Female</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Tariff</i> $\times$ <i>Female</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>PNTR</i> $\times$ <i>Female</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>FDI</i> $\times$ <i>Genealogy</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Tariff</i> $\times$ <i>Genealogy</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>PNTR</i> $\times$ <i>Genealogy</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Age16</i> $\times$ <i>Female</i>	Absorbed by <i>Province-Cohort-Gender Fixed Effects</i>
<i>Age16</i> $\times$ <i>Genealogy</i>	Absorbed by <i>Genealogy</i> $\times$ <i>Cohorts</i>
<i>IndivChar</i> $\times$ <i>Female</i>	Interactions of Han ethnicity or Hukou status with the female indicator.
<i>Genealogy</i> $\times$ <i>Female</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>RegionChar</i> $\times$ <i>Female</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
Triple interaction terms	
<i>FDI</i> $\times$ <i>Age16</i> $\times$ <i>Female</i>	Triple interactions of FDI deregulation with age and the female indicator.
<i>Tariff</i> $\times$ <i>Age16</i> $\times$ <i>Female</i>	Triple interactions of import tariff reduction with age and the female indicator.
<i>PNTR</i> $\times$ <i>Age16</i> $\times$ <i>Female</i>	Triple interactions of PNTR intensity with age and the female indicator.
<i>FDI</i> $\times$ <i>Age16</i> $\times$ <i>Genealogy</i>	Triple interactions of FDI deregulation with age and the genealogy indicator.
<i>Tariff</i> $\times$ <i>Age16</i> $\times$ <i>Genealogy</i>	Triple interactions of import tariff reduction with age and the genealogy indicator.
<i>PNTR</i> $\times$ <i>Age16</i> $\times$ <i>Genealogy</i>	Triple interactions of PNTR intensity with age and the genealogy indicator.
<i>FDI</i> $\times$ <i>Female</i> $\times$ <i>Genealogy</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Tariff</i> $\times$ <i>Female</i> $\times$ <i>Genealogy</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>PNTR</i> $\times$ <i>Female</i> $\times$ <i>Genealogy</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Genealogy</i> $\times$ <i>Cohorts</i> $\times$ <i>Female</i>	Controlling for gender-differential trends in initial genealogy density across prefectures.
<i>RegionChar</i> $\times$ <i>Cohorts</i> $\times$ <i>Female</i>	Controlling for gender-differential trends in initial regional characteristics. ^a
Quadruple interaction terms	
<i>FDI</i> $\times$ <i>Age16</i> $\times$ <i>Female</i> $\times$ <i>Genealogy</i>	Quadruple interactions of FDI deregulation with age, female and genealogy indicator.
<i>Tariff</i> $\times$ <i>Age16</i> $\times$ <i>Female</i> $\times$ <i>Genealogy</i>	Quadruple interactions of import tariff reduction with age, female and genealogy indicator.
<i>PNTR</i> $\times$ <i>Age16</i> $\times$ <i>Female</i> $\times$ <i>Genealogy</i>	Quadruple interactions of PNTR intensity with age, female and genealogy indicator.

^a Pre-determined regional characteristics include average years of schooling and the agricultural- and manufacturing-employment shares for people aged 16–60 in the 1990 China Population Census.

Table A5: Detailed terms in Column (2)–(4) of Table A3

Specific term	Description
Fixed effects and trends	
<i>Prefecture-Gender Fixed Effects</i>	Capturing time-invariant characteristics that vary by prefecture and gender.
<i>Province-Cohort-Gender Fixed Effects</i>	Capturing province-by-gender-specific flexible trends.
<i>RegionChar</i> $\times$ <i>Cohorts</i>	Controlling for flexible trends in initial regional characteristics. ^a
<i>Genealogy</i> $\times$ <i>Cohorts</i>	Controlling for flexible trends in initial genealogy density across prefectures.
<i>Confounder</i> $\times$ <i>Cohorts</i>	Controlling for flexible trends in each potential confounder across prefectures. ^b
Level terms	
<i>FDI</i>	Prefecture-level FDI deregulation, absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Tariff</i>	Prefecture-level import tariff reduction, absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>PNTR</i>	Prefecture-level Column 2 tariff rate, absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Age16</i>	Absorbed by <i>Province-Cohort-Gender Fixed Effects</i>
<i>Female</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Genealogy</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Confounder</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>RegionChar</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>IndivChar</i>	Individual characteristics, i.e. indicators of Han ethnicity and Hukou status.
Two-way interaction terms	
<i>FDI</i> $\times$ <i>Age16</i>	The interaction terms of FDI deregulation with age indicator.
<i>Tariff</i> $\times$ <i>Age16</i>	The interaction terms of import tariff reduction with age indicator.
<i>PNTR</i> $\times$ <i>Age16</i>	The interaction terms of PNTR intensity with age indicator.
<i>FDI</i> $\times$ <i>Female</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Tariff</i> $\times$ <i>Female</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>PNTR</i> $\times$ <i>Female</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>FDI</i> $\times$ <i>Genealogy</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Tariff</i> $\times$ <i>Genealogy</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>PNTR</i> $\times$ <i>Genealogy</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>FDI</i> $\times$ <i>Confounder</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Tariff</i> $\times$ <i>Confounder</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>PNTR</i> $\times$ <i>Confounder</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Age16</i> $\times$ <i>Female</i>	Absorbed by <i>Province-Cohort-Gender Fixed Effects</i>
<i>Age16</i> $\times$ <i>Genealogy</i>	Absorbed by <i>Genealogy</i> $\times$ <i>Cohorts</i>
<i>Age16</i> $\times$ <i>Confounder</i>	Absorbed by <i>Confounder</i> $\times$ <i>Cohorts</i>
<i>IndivChar</i> $\times$ <i>Female</i>	Interactions of Han ethnicity or Hukou status with the female indicator.
<i>Genealogy</i> $\times$ <i>Female</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>RegionChar</i> $\times$ <i>Female</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Confounder</i> $\times$ <i>Female</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
Triple interaction terms	
<i>FDI</i> $\times$ <i>Age16</i> $\times$ <i>Female</i>	Triple interactions of FDI deregulation with age and the female indicator.
<i>Tariff</i> $\times$ <i>Age16</i> $\times$ <i>Female</i>	Triple interactions of import tariff reduction with age and the female indicator.
<i>PNTR</i> $\times$ <i>Age16</i> $\times$ <i>Female</i>	Triple interactions of PNTR intensity with age and the female indicator.
<i>FDI</i> $\times$ <i>Age16</i> $\times$ <i>Genealogy</i>	Triple interactions of FDI deregulation with age and the genealogy indicator.
<i>Tariff</i> $\times$ <i>Age16</i> $\times$ <i>Genealogy</i>	Triple interactions of import tariff reduction with age and the genealogy indicator.
<i>PNTR</i> $\times$ <i>Age16</i> $\times$ <i>Genealogy</i>	Triple interactions of PNTR intensity with age and the genealogy indicator.
<i>FDI</i> $\times$ <i>Age16</i> $\times$ <i>Confounder</i>	Triple interactions of FDI deregulation with age and each potential confounder. ^b
<i>Tariff</i> $\times$ <i>Age16</i> $\times$ <i>Confounder</i>	Triple interactions of import tariff reduction with age and each potential confounder. ^b
<i>PNTR</i> $\times$ <i>Age16</i> $\times$ <i>Confounder</i>	Triple interactions of PNTR intensity with age and each potential confounder. ^b
<i>FDI</i> $\times$ <i>Female</i> $\times$ <i>Genealogy</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>Tariff</i> $\times$ <i>Female</i> $\times$ <i>Genealogy</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
<i>PNTR</i> $\times$ <i>Female</i> $\times$ <i>Genealogy</i>	Absorbed by <i>Prefecture-Gender Fixed Effects</i>

Table A5 – Continued

Specific term	Description
$FDI \times Female \times Confounder$	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
$Tariff \times Female \times Confounder$	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
$PNTR \times Female \times Confounder$	Absorbed by <i>Prefecture-Gender Fixed Effects</i>
$Genealogy \times Cohorts \times Female$	Controlling for gender-differential trends in initial genealogy density across prefectures.
$RegionChar \times Cohorts \times Female$	Controlling for gender-differential trends in initial regional characteristics. ^a
$Confounder \times Cohorts \times Female$	Controlling for gender-differential trends in each potential confounder. ^b
Quadruple interaction terms	
$FDI \times Age16 \times Female \times Genealogy$	Quadruple interactions of FDI deregulation with age, female and genealogy indicator.
$Tariff \times Age16 \times Female \times Genealogy$	Quadruple interactions of import tariff reduction with age, female and genealogy indicator.
$PNTR \times Age16 \times Female \times Genealogy$	Quadruple interactions of PNTR intensity with age, female and genealogy indicator.
$FDI \times Age16 \times Female \times Confounder$	Quadruple interactions of FDI deregulation with age, female and each potential confounder. ^b
$Tariff \times Age16 \times Female \times Confounder$	Quadruple interactions of import tariff reduction with age, female and each potential confounder. ^b
$PNTR \times Age16 \times Female \times Confounder$	Quadruple interactions of PNTR intensity with age, female and each potential confounder. ^b

^a Pre-determined regional characteristics include average years of schooling and the agricultural- and manufacturing-employment shares for people aged 16–60 in the 1990 China Population Census.

^b Confounders comprise the pre-existing gender gap for the 1970–1980 cohorts, night-light strength in 2000, and the employment share of private manufacturing enterprises in 2000.

A.3 Industry-Level Relative Importance of Female-Oriented Skills

In this section, we outline our methodology for constructing an industry-level relative importance of female-oriented skill intensity that integrates information on gender-specific skill requirements across industries. To develop this measure, we utilize the Occupational Information Network (O*NET) database, which provides comprehensive data on both the importance and required proficiency levels of various skills and abilities across occupations.²

We begin by identifying two categories of skills that capture gender-specific comparative advantage. Following Li (2021), we classify eight female-oriented skills into two domains: (i) **verbal and written communication**: oral comprehension, oral expression, written comprehension, and written expression; and (ii) **social interaction**: negotiation, persuasion, service orientation, and coordination. These skill categories are detailed in Panel A of Table A6. Together, these dimensions represent domains in which women systematically demonstrate superior performance relative to men.³

In parallel, we define male-oriented skills as those emphasizing spatial cognitive ability and physical strength, areas where men exhibit comparative advantage. Specifically, we categorize skills in which males hold a comparative advantage into two domains: (i)

²The importance score quantifies the degree to which a particular skill or ability is essential for job performance in a given occupation, while the level score indicates the proficiency required. Both metrics are scaled from 0 to 100 and are determined by domain specialists and practitioners within the respective occupational fields.

³A substantial body of biological and psychological research documents female advantages in verbal and written communication (Kramer et al., 1997; Crossley, D’arcy, and Rawson, 1997; Wilcox, Maxey, and Herbst, 1992) as well as in social cognition and empathy (Hall, 1978; Chapman et al., 2006; Feingold, 1994).

cognitive-spatial abilities: spatial orientation and visualization; and (ii) **physical skills:** explosive strength, static strength, stamina, trunk strength, gross body coordination, and extent flexibility, as detailed in Panel B of Table A6.

Table A6: Gender-oriented skills

Skill Category	Specific Skills
Panel A. Female-oriented skills	
Verbal and Written Language Abilities	Oral Comprehension
	Oral Expression
	Written Comprehension
	Written Expression
Social Skills	Negotiation
	Persuasion
	Service Orientation
	Coordination
Panel B. Male-oriented skills	
Cognitive Spatial Abilities	Spatial Orientation
	Visualization
Physical Skills	Explosive Strength
	Static Strength
	Stamina
	Trunk Strength
	Gross Body Coordination
	Extent Flexibility

Note: This table shows skill categories alongside their corresponding specific skills, which are utilized to construct occupation-level or industry-level intensities of female- and male-oriented skills, respectively. The classification of gender-oriented skills follows the framework established by Li (2021).

Using a shift-share approach, we calculate the weighted average importance scores for both female-oriented and the complement of male-oriented skills across industries. The construction proceeds as follows:

$$t_{q,k}^g = \sum_{o=1} s_{k,o} \times t_{q,o}^g, \quad (2)$$

where k indexes industries, o indexes occupations, q indexes the specific skill, and g denotes the skill category (female- or male-oriented) to which skill q belongs. The term $s_{k,o}$ denotes the employment share of occupation o within industry k , derived from the Occupational Employment Statistics (OES) Survey conducted by the U.S. Bureau of Labor Statistics.⁴ The term $t_{q,o}^g$ denotes the importance score of gender-oriented skill q within skill category g

⁴The OES Survey provides comprehensive employment data on occupations and industries based on the 7-digit Standard Occupational Classification system, with a concordance to 3-digit Standard Industrial Classification (SIC) industries. We employ employment shares from 2000, a period preceding China's WTO accession.

for occupation o . Following the methodology in Li (2021), we compute the complement of each male-oriented skill score by subtracting each raw value from 100, so that higher values correspond to occupations with lower reliance on male-oriented skills. The resulting $t_{q,k}^g$ represents the importance score of gender-oriented skill q within skill category g for industry k .

In total, we have eight female-oriented skill importance scores (i.e., $t_{q,k}^{female}$) and the eight importance scores of male-oriented skill complements at the industry level (i.e., $t_{q,k}^{male}$). We apply principal component analysis (PCA) on the 16 scores, and denote the first principal component as a composite index on the relative importance of female-oriented skills in industry k (i.e., $SkillInt_k^F$). Higher values of relative female-oriented skill importance indicate industries with greater reliance on skills aligned with female comparative advantage and lower utilization of skills in which males demonstrate dominant comparative advantage, and vice versa.

The industry-level measures of relative female-oriented skill importance are initially coded using the 3-digit U.S. SIC system. We then map these measures to the International Standard Industrial Classification and subsequently to the CIC system to facilitate integration with Chinese census data.

A.4 Occupational Rankings by Female-Skill Intensity

To provide further insights into occupational variation in female-skill intensity. We construct occupational female-skill intensity measures akin to the methodology outlined in Section A.3. Specifically, we apply PCA to 16 occupational skill importance scores from O*NET: 8 female-oriented skills (denoted $t_{q,o}^{female}$) and 8 complementary male-oriented skills (denoted $t_{q,o}^{male}$). The first principal component yields a composite index, $SkillInt_o^F$, which captures the relative importance of female-specific skills within each occupation. Table A7 presents rankings of occupations based on these occupation-level relative female-skill intensity.

Panel A of Table A7 lists the top 10 and bottom 10 occupations by female-skill intensity across all 704 occupations. Occupations with the highest female-skill intensity are predominantly service-oriented and professional positions requiring substantial interpersonal communication and coordination skills, such as Lawyers, Medical and Health Services Managers, and Public Relations Managers and Specialists. In contrast, occupations with the lowest female-skill intensity are concentrated in physically demanding trades requiring explo-

Table A7: Ranking of occupations by female-oriented skill intensity

Highest Ten		Lowest Ten	
Rank	Occupations	Rank	Occupations
Panel A. All Occupations			
1	Lawyers	1	Reinforcing Iron and Rebar Workers
2	Clinical, Counseling, and School Psychologists	2	Fallers
3	Education Administrators, Postsecondary	3	Helpers–Masons and Tile Setters
4	Medical and Health Services Managers	4	Foundry Mold and Coremakers
5	Arbitrators, Mediators, and Conciliators	5	Terrazzo Workers and Finishers
6	Agents and Business Managers of Artists, Performers, and Athletes	6	Landscaping and Groundskeeping Workers
7	Clergy	7	Dancers
8	Public Relations Managers	8	Brickmasons and Blockmasons
9	Public Relations Specialists	9	Roof Bolters, Mining
10	Sales Engineers	10	Drywall and Ceiling Tile Installers
Panel B. Weighted by Occupational Employment Share in FDI-Reformed Industries			
1	Retail Salespersons	1	Team Assemblers
2	General and Operations Managers	2	Auto Service Techs and Mechanics
3	Supervisors of Production Workers	3	Cutting and Press Machine Operators
4	Wholesale and Manufacturing Sales Reps	4	Laborers and Material Movers, Hand
5	Customer Service Representatives	5	Cleaners of Vehicles and Equipment
6	Bookkeeping, Accounting, and Auditing Clerks	6	Welders, Cutters, Solderers, and Brazers
7	Office Clerks, General	7	Machinists
8	Executive Secretaries and Assistants	8	Helpers–Production Workers
9	Secretaries, Except Legal/Medical	9	Electrical and Electronic Assemblers
10	Purchasing Agents, Except Farm Products	10	Packers and Packagers, Hand

Note: This table presents the occupations with the highest and lowest female-oriented skill intensity. Occupational classifications and corresponding titles follow the SOC system. A PCA approach is employed to construct occupation-level measures of female-skill intensity. Panel A presents the rankings of all 704 occupations. Panel B shows the rankings of occupations based on female-skill intensity weighted by occupational employment shares in industries affected by FDI shocks. We exclude occupations with zero employment in FDI-liberalized industries, yielding 341 occupations.

sive strength and spatial abilities, such as Reinforcing Iron and Rebar Workers, Fallers, and Terrazzo Workers and Finishers.

Nonetheless, occupations with the highest female-skill intensity may not be prevalent in industries most affected by FDI deregulation. For instance, Education Administrators and Clinical and Counseling Psychologists are concentrated in public education and healthcare sectors, which are unlikely to constitute a large share of FDI-liberalized industries. This is particularly relevant given that FDI deregulation in China primarily targeted manufacturing sectors rather than service industries.

Hence, in order to identify female-skill-intensive occupations relevant to FDI liberalization industries, we weight occupational female-skill intensity by employment shares in industries experiencing FDI deregulation. Specifically, we calculate the weighted female-skill

intensity as follows:

$$SkillIntWeight_o^{F,FDI} = SkillInt_o^F \times s_o^{FDI}, \quad (3)$$

where $s_o^{FDI} = Emp_o^{FDI}/Emp^{FDI}$ denotes the employment share of occupation o in FDI-reformed industries. The term Emp_o^{FDI} represents the number of workers in occupation o employed in FDI-deregulated industries, and Emp^{FDI} is total employment across all FDI-deregulated industries. The term $SkillInt_o^F$ represents occupation o 's female-skill intensity from our PCA. The weighted measure $SkillIntWeight_o^{F,FDI}$ thus captures variation in female-skill intensity among occupations prevalent in industries exposed to FDI liberalization. We exclude occupations with zero employment in FDI-deregulated industries (i.e., $s_o^{FDI} = 0$).

Panel B of Table A7 ranks occupations by this weighted female-skill intensity. The top-ranked occupations are concentrated in managerial, supervisory, and administrative positions, such as General and Operations Managers, Supervisors of Production Workers, Office Clerks, and Executive Secretaries and Assistants. These occupations require substantial communication, coordination, and interpersonal skills—areas where women hold comparative advantages. Consistent with the unconditional rankings in Panel A, the bottom-ranked occupations are concentrated in positions requiring spatial abilities (e.g., spatial orientation and visualization) or physical strength (e.g., explosive and static strength), such as Cutting and Press Machine Operators and Laborers and Material Movers.

Appendix B Supplementary Figures and Tables

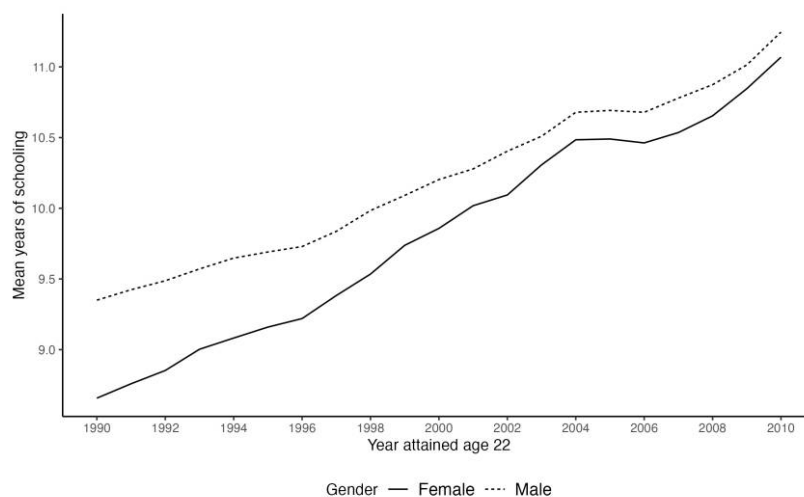


Figure B1: Trends of gender gap in educational attainment

Note: This figure displays mean years of schooling by gender for cohorts reaching age 22 between 1990 and 2010. The horizontal axis denotes the year in which each cohort reaches age 22. The vertical axis denotes mean years of schooling. The solid line represents females and the dashed line represents males. Data are from the 2010 China Population Census.

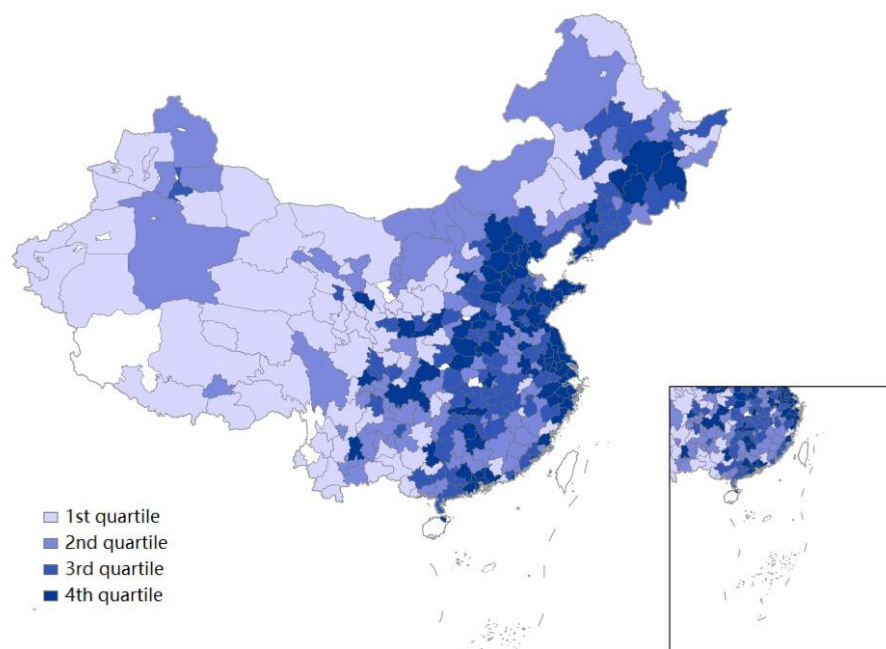


Figure B2: The spatial distribution of FDI deregulation intensity

Notes: This figure displays the spatial distribution of regional FDI deregulation intensity. Each block represents a prefecture-level city in China. Cities are classified into four quartiles based on FDI deregulation intensity, with darker shading indicating higher intensity.

Table B1: Impacts of FDI deregulation on FDI activities

	lnFDI		shFDI	
	(1)	(2)	(3)	(4)
$FDI \times Post$	0.225* (0.115)	0.327* (0.170)	0.052** (0.023)	0.056** (0.028)
Observations	3110	608	3110	608
R-squared	0.90	0.80	0.92	0.86
Weight	Yes			
Two Periods (00 & 05)	No	Yes	No	Yes
Controls	Yes			
Fixed Effects	Prefecture, Wave, Province-Post			

Notes: Weight is the city's population. Two periods' results are obtained using only the years 2000 and 2005 to be consistent with the census data we use. Other columns include all years from 1998–2007. Columns (1) and (2) use the log number of foreign-invested enterprises (FIEs) at the prefecture level as the dependent variable. Columns (3) and (4) use the FIE share, defined as the ratio of FIEs to total manufacturing firms within each prefecture. Controls include interactions between the post dummy and a set of pre-determined prefecture characteristics. The pre-determined prefecture characteristics include the average years of schooling, the share of agricultural employment, and the share of manufacturing employment for people aged 16–60 in the 1990 China population census. Standard errors in parentheses are clustered by prefecture. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B2: Migration rate

Dependent variable	In-migration rate			Out-migration rate		
	Both (1)	Male (2)	Female (3)	Both (4)	Male (5)	Female (6)
$FDI \times Post$	-0.010 (0.011)	-0.007 (0.012)	-0.013 (0.011)	0.008 (0.023)	0.008 (0.024)	0.008 (0.022)
Observations	982	982	982	982	982	982
R-squared	0.98	0.97	0.98	0.73	0.72	0.73
Controls	Yes					
Fixed Effects	Prefecture, Province-by-year					

Notes: The sample includes prefecture-level data calculated from the 2000, 2005, and 2010 waves of the China population censuses. In columns (1)–(3), the dependent variable is the in-migration rate. In columns (4)–(6), the dependent variable is the out-migration rate. Control variables include the share of Han, the share of males, tariff and uncertainty shocks, and interaction between wave dummies and a set of pre-determined prefecture characteristics. The tariff and uncertainty shocks include interactions of initial import tariff and column 2 tariff levels with the post dummy. The pre-determined prefecture characteristics include the average schooling years, the share of agriculture employment, and the share of manufacturing employment for people aged 16–60 in the 1990 China population census. Standard errors in parentheses are clustered by prefecture. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B3: Middle school and college attendance

Dependent variable	Middle-school attendance			College attendance		
	Male (1)	Female (2)	Full (3)	Male (4)	Female (5)	Full (6)
$FDI \times AgeKey$	0.010 (0.008)	0.004 (0.010)	0.010 (0.008)	0.020 (0.017)	-0.002 (0.011)	0.020 (0.017)
$FDI \times AgeKey \times Female$			-0.007 (0.008)			-0.022 (0.020)
Controls	Yes					
Fixed Effects	Prefecture, Province-by-cohort					
Observations	405,714	405,166	810,880	304,554	317,639	622,193
R-squared	0.13	0.14	0.13	0.29	0.33	0.31

Notes: In columns (1)–(3), the dependent variable is an indicator of middle-school attendance. The sample includes people born in 1986–1991 in the 2005 and 2010 waves of the China population (mini-)census. They were 11–16 years old in 2002, when the FDI regulations were relaxed. In columns (4)–(6), the dependent variable is an indicator of college attendance. The sample includes people born in 1980–1985 in the 2005 and 2010 waves of the China population (mini-)census. They were 17–22 years old in 2002, when the FDI regulations were relaxed. Controls are the set of covariates defined in Table 2. All control variables are interacted with the female dummy to allow for gender-specific effects. Standard errors in parentheses are clustered by prefecture. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B4: Statistical inference: Alternative cluster-level

Dependent variable	High-school attendance				
Cluster level	Province	Employment similarity	Employment similarity outside prov	Prov + Employment similarity	Prov + Employment similarity outside prov
	(1)	(2)	(3)	(4)	(5)
$FDI \times Age16 \times Female$	0.056*** (0.017)	0.056*** (0.014)	0.056*** (0.010)	0.056*** (0.016)	0.056*** (0.011)
$FDI \times Age16$	-0.038** (0.014)	-0.038*** (0.010)	-0.038*** (0.008)	-0.038*** (0.013)	-0.038*** (0.010)
Controls	Yes				
Fixed effects	Prefecture, Province-by-cohort				
Observations	698,407				
R-squared	0.27				

Note: The table reports the baseline estimates from Column (1) of Table 2 with standard errors clustered at several alternative levels. Column (1) presents results clustered at the provincial level. Columns (2) and (3) present results with standard errors clustered at the level of employment-similarity groups and at the level of employment-similarity groups defined outside the province, respectively. Columns (4) and (5) employ two-way clustering: at the province and employment-similarity group levels (Column 4), and at the province and outside-of-province employment-similarity group levels (Column 5). Controls are identical to those used in the baseline specification in Table 2. All control variables are interacted with the female dummy to allow for gender-specific effects. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B5: Balance tests on FDI shocks

Dep Var:	Indicator on FDI deregulation		
	(1)	(2)	(3)
HHI index	-0.383 (0.294)	-0.306 (0.325)	-0.429 (0.372)
New product share	0.404 (0.253)	0.185 (0.309)	0.286 (0.429)
Export share	0.050 (0.127)	0.111 (0.142)	0.054 (0.220)
SOE share	-0.536*** (0.175)	-0.451* (0.239)	-0.280 (0.249)
Average firm age (log)	0.242*** (0.085)	0.186 (0.117)	0.118 (0.130)
Firm number (log)	-0.060** (0.025)	-0.055** (0.026)	-0.067** (0.032)
2-digit industry FE	Yes		
3-digit industry FE	Yes		
Number of industries (4-digit)	436	436	369
R-squared	0.05	0.13	0.34

Notes: This table shows the regression results of the indicator on FDI liberation on industry characteristics in 2000. Standard errors in parentheses are clustered by 3-digit industries. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B6: Industry-level regressions

Dependent variable:	Cohort difference of the gender gap in high-school attendance rate			
	(1)	(2)	(3)	(4)
FDI deregulation	-0.066*** (0.022)	-0.064*** (0.022)	-0.057** (0.023)	-0.074** (0.033)
HHI index		-0.034** (0.014)	-0.031** (0.014)	-0.027 (0.017)
New product share		-0.003 (0.004)	0.003 (0.006)	0.009 (0.008)
Export share		-0.001 (0.003)	0.001 (0.003)	0.001 (0.004)
SOE share		0.006 (0.004)	0.008 (0.006)	0.008 (0.009)
Average firm age (log)		0.001 (0.002)	0.001 (0.002)	0.002 (0.004)
Firm number (log)		-0.001** (0.001)	-0.001 (0.001)	-0.001 (0.001)
2-digit industry FE			Yes	
3-digit industry FE				Yes
Number of industries (4-digit)	436	436	436	369

Notes: This table shows robustness tests at the industry level. Standard errors in parentheses are clustered by 3-digit industries. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B7: Robustness to incomplete shares

Dependent variable Sample:	High-school attendance		
	Male	Female	Full
	(1)	(2)	(3)
<i>Panel A. Benchmark</i>			
$FDI \times Age16$	-0.038** (0.015)	0.018 (0.016)	-0.038** (0.015)
$FDI \times Age16 \times Female$			0.056*** (0.017)
Observations	343,087	355,320	698,407
R-squared	0.25	0.28	0.27
<i>Panel B. Control for the sum of shares across prefectures</i>			
$FDI \times Age16$	-0.043*** (0.015)	0.021 (0.016)	-0.043*** (0.015)
$FDI \times Age16 \times Female$			0.064*** (0.018)
Observations	343,087	355,320	698,407
R-squared	0.25	0.28	0.27
Contemporaneous policy controls		Yes	
Initial conditions # cohorts		Yes	
Fixed effects	Prefecture, Province-by-cohort		

Notes: This table presents robustness checks addressing incomplete share bias following [Borusyak, Hull, and Jaravel \(2022\)](#). Panel A replicates baseline estimates from Table 2. Panel B augments the baseline specification with controls for the sum of exposure shares. Columns (1) and (2) report estimates for male and female subsamples, respectively, while Column (3) pools both genders. In Panel B, Columns (1) and (2) include two-way interaction terms between the sum of exposure shares across prefectures and cohort dummies. Column (3) additionally includes a triple interaction term between prefecture-level sum of exposure shares, cohort dummies, and a female indicator. All specifications include the same control variables as in Table 2. All control variables are interacted with the female dummy to allow for gender-specific effects. Standard errors in parentheses are clustered by prefecture. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B8: Validity test for recentered IV

Dependent variable:	Recentered FDI Shock		
	(1)	(2)	(3)
Distance to Beijing	0.014 (0.042)		0.007 (0.039)
Latitude	-0.001 (0.003)		-0.000 (0.003)
Longitude	0.003 (0.003)		0.002 (0.003)
Expected FDI Shock		0.046 (0.030)	0.045 (0.030)
Fixed effects		Province	
Observations	328	328	328
R-squared	0.10	0.13	0.13

Notes: This table shows validity tests for recentered instrument at prefecture level. Standard errors in parentheses are clustered by the province level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B9: Recentered estimates

Dependent variable	High-school attendance	
Methods	Recentered IV	Controlled OLS
	(1)	(2)
$FDI_{rc} \times Age16 \times Female$	0.060** (0.024)	
$FDI_{rc} \times Age16$	-0.038* (0.020)	
$FDI \times Age16 \times Female$		0.067*** (0.022)
$FDI \times Age16$		-0.052*** (0.018)
$FDI_{exp} \times Age16 \times Female$		-0.022 (0.029)
$FDI_{exp} \times Age16$		0.027 (0.026)
Controls	Yes	
Fixed effects	Prefecture, Province-by-cohort	
Observations	698,407	698,407
R-squared	0.27	0.27

Note: The sample comprises individuals born between 1983 and 1988, drawn from the 2005 and 2010 waves of the China population (mini-)census. These individuals were aged 14–19 in 2002 when FDI regulations were relaxed. The variables FDI_{rc} and FDI_{exp} denote the recentered and expected instruments, respectively, which are constructed by permuting the FDI deregulation indicator within 3-digit industry classifications. Control variables are consistent with those in Table 2. All control variables are interacted with the female dummy to allow for gender-specific effects. Standard errors in parentheses are clustered at prefecture level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B10: Worker composition of reformed and unreformed industries

Sample	Reformed (84 industries)		Unreformed (89 industries)		<i>t</i> test	
	Mean	S.D.	Mean	S.D.	Col.(1)-(3)	<i>p</i> value
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Both males and females						
Male	0.60	0.49	0.48	0.50	0.12***	[0.000]
Age	33.44	10.68	32.02	10.53	1.42***	[0.000]
Schooling years	9.56	2.76	9.35	2.63	0.21***	[0.000]
Observations	451,203		390,800		842,003	
Panel B. Males						
Age	34.66	11.09	34.18	11.05	0.49***	[0.000]
Schooling years	9.68	2.72	9.70	2.65	-0.02**	[0.021]
Observations	272,453		187,734		460,187	
Panel C. Females						
Age	31.58	9.73	30.03	9.60	1.55***	[0.000]
Schooling years	9.37	2.80	9.02	2.56	0.35***	[0.000]
Observations	178,750		203,066		381,816	

Notes: The data source is the 2000 China population census. We define a 3-digit CIC industry as a reformed industry if at least one of the enclosed 4-digit CIC industries was deregulated in 2002 and zero otherwise.

Table B11: Labor-market outcomes in the 2010 census

Sample	Full		Male		Female	
	Mean	S.D.	Mean	S.D.	Mean	S.D.
	(1)	(2)	(3)	(4)	(5)	(6)
Age	24.136	1.739	24.135	1.741	24.138	1.737
Employment status	0.805	0.396	0.861	0.345	0.749	0.433
Labor-force participation	0.849	0.358	0.901	0.298	0.799	0.401
White-collar job	0.142	0.349	0.130	0.336	0.156	0.363
Observations	495,679		244,651		251,028	

Notes: The data source is the 2010 wave of the China population census. We include people born in 1983–1988, who were 14–19 years old at the relaxation of the FDI regulations in 2002. Most of the sample people have had completed education by 2010.

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